

Single-cell foundation models represent genes in context, but reveal no context-specific gene function beyond co-expression

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Abstract

Background: Single-cell foundation models build, at every transformer layer, a representation of each gene that depends on the other genes in the cell. Whether this contextual representation shows how a gene's role differs between cell types, the single-cell counterpart of word sense in language models, has not been tested.

Results: We measured contextual gene representations in MaxToki (217M and 1B), scGPT and STATE on identical Tabula Sapiens cells from 12 cell types, and in MaxToki-217M also on independent bone-marrow data. In every model a gene's representation changes with cell type in a gene-specific way: the change agrees between two independent halves of the cells, with a cosine 0.74 to 0.87 higher than for a randomly paired gene. It is not explained by the gene's rank in the cell and replicates on the bone-marrow data. Untrained versions keep 36–88% of it, and a representation built from expression alone also shows it (+0.321). In MaxToki-217M the change follows functional axes such as nuclear versus surface genes ($z = +13.3$ against random axes; on this axis STATE shows less and scGPT little). In all four models it follows co-expressed gene sets. None of three expression-only representations shows either kind of organisation. In no model is the functional organisation significantly stronger than that of gene sets matched on co-expression (smallest $p = 0.07$), although this test detects only large differences. Adding the nuclear/surface direction to half of a cell's genes shifts MaxToki's predictions for the other half more than random directions from the same genes; as the pushes exceed natural changes, this shows the model can read the direction, not that it uses it. In both MaxToki models, genes that change more gain slightly more from their own context in next-gene prediction. Two direct tests for context-specific gene function were negative in all four models.

Conclusions: Single-cell foundation models build real contextual gene representations, organised by co-expression. We found no context-specific gene function beyond co-expression, though our tests could detect only clear effects. Models trained on more than expression were not tested. Functional organisation separates some architectures but does not rank them reliably.

Keywords: Single-cell foundation models; Contextual gene representations; Mechanistic interpretability; Gene co-expression; Gene embeddings; Transformers; Single-cell RNA sequencing; Model evaluation

Background

A recurring theme in the interpretability of large language models is that a network trained on a narrow predictive task builds internal representations of structure it was never taught. Sequence models trained on Othello and chess build internal representations of the board and of hidden variables such as player skill (Li et al. 2023; Nanda et al. 2023; Karvonen 2024). Language models encode space and time along single directions that a linear probe can read (Gurnee and Tegmark 2024). In the best cases the model also uses

these representations to make its predictions. One tests this by changing the representation and measuring the effect on the model’s output (Turner et al. 2023; Zou et al. 2023; Park et al. 2024).

A closely related line concerns contextual representations. Contextual word embeddings assign the same word a different vector in every sentence, and the geometry of those vectors encodes word sense: “bank” near a river and “bank” in finance occupy separable regions (Peters et al. 2018; Ethayarajh 2019; Coenen et al. 2019). These measurements need care, because representation spaces are strongly anisotropic (vectors cluster in a narrow cone rather than spreading evenly, so any two look similar) and a handful of “rogue” dimensions can dominate cosine similarity (Ethayarajh 2019; Mu et al. 2018; Cai et al. 2021; Timkey and van Schijndel 2021).

Single-cell foundation models (SCFMs) offer a clean biological version of the same question (Theodoris et al. 2023; Cui et al. 2024; Hao et al. 2024; Yang et al. 2022). They treat each gene as a token and each cell as a set or sequence of genes, either ordered by expression (Geneformer) or paired with binned or continuous expression values (scGPT, scBERT, scFoundation), and are trained to predict masked or next genes or values, in effect learning which genes tend to be active together. Their gene token embeddings (one fixed vector per gene) have been studied (Cui et al. 2024; Kendiukhov 2026f); their contextual gene representations (a gene’s vector inside a particular cell) are used as features downstream (Hao et al. 2024; Bian et al. 2024). Their geometry has been mapped when averaged over cells or within one tissue (Related work), but how a single gene’s vector changes from one cell type to another has not been measured. A model could use this vector to represent that a gene plays a different role in a macrophage than in a T cell. That would be the biological counterpart of word sense. The “cell as a token” framing points to this question but does not test it (Gilpin 2025).

Related work. Our methods descend directly from the language-model literature on contextualised embeddings. Ethayarajh (Ethayarajh 2019) showed that contextual representations are anisotropic in all non-input layers, that same-word self-similarity falls in upper layers, and that every cosine must be corrected against an anisotropy baseline. Timkey and van Schijndel (Timkey and van Schijndel 2021) showed that a few rogue dimensions dominate similarity and that a scalar correction is not enough; Mu et al. (Mu et al. 2018) and Cai et al. (Cai et al. 2021) give the standard postprocessing and the local-isotropy picture. Coenen et al. (Coenen et al. 2019) and Garí Soler and Apidianaki (Garí Soler and Apidianaki 2021) show that word senses form separable clusters and that a word’s degree of polysemy is itself readable. We import both the measurements and the controls. From the literature on latent variables that models use causally (Li et al. 2023; Nanda et al. 2023; Karvonen 2024; Gurnee and Tegmark 2024) we adopt one rule: a probe shows only correlation, so a causal claim needs an intervention compared against a matched control. We also use the steering methods of this literature (Turner et al. 2023; Zou et al. 2023). Templeton et al. did the same at large scale, extracting interpretable feature directions from a production model and steering it with them (Templeton et al. 2024).

Geneformer (Theodoris et al. 2023), scGPT (Cui et al. 2024), scFoundation (Hao et al. 2024), scBERT (Yang et al. 2022), Universal Cell Embeddings (UCE) (Rosen et al. 2026) and STATE (Adduri et al. 2026) differ in tokenisation (rank-ordered gene identity, binned or continuous expression values, or gene tokens derived from protein-sequence embeddings), and the review of Bian et al. (Bian et al. 2024) defines the “contextual gene embedding” that our work dissects. MaxToki (Gómez Ortega et al. 2026), the primary model here, is a Llama-architecture, Geneformer-lineage model at 217M and 1B scales. Two earlier analyses come closest to a “context changes the gene” claim. scGPT reports cell-state-specific gene networks read from attention (Cui et al. 2024), but as a pairwise network, not as a property of the representation vector, and without a decomposition or a null. Geneformer’s in silico deletion measures how the embeddings of the other genes shift when one gene is removed from a cell (Theodoris et al. 2023), which is a change of context by one gene. Neither characterises how a gene’s representation differs between natural cell types. The geometry of gene representations in these models has been mapped, both for static gene embeddings (Kendiukhov 2026f) and for layer-wise hidden states averaged over cells, including their spectral structure (Kendiukhov 2026b) and the topological and geometric structure they do and do not encode, per tissue (Kendiukhov 2026g). Residual-stream gene vectors built separately in several tissues carry a small, tissue-dependent amount of transcription-factor-target signal beyond expression covariates (Kendiukhov 2026c). That test used TRRUST edges, which are the same in every tissue, so it did not ask whether a gene’s role differs between tissues.

A critical literature finds that SCFMs often fail to beat simple baselines: on zero-shot tasks (Kedzierska et al. 2025), under deeper evaluation (Boiarsky et al. 2024), on perturbation prediction (Ahlmann-Eltze et al. 2025; Bendidi et al. 2024), and across utilities (Liu et al. 2026). Sparse autoencoders find organised biological knowledge but minimal regulatory logic in these models (Kendiukhov 2026d), and edge scores read from SCFM attention add nothing beyond simple gene-level statistics for predicting which genes respond to a perturbation (Kendiukhov 2026e). Co-expression is the canonical guilt-by-association signal (van Dam et al. 2018), and some gene-embedding methods, such as Gene2vec, are built directly on it (Du et al. 2019); others build gene embeddings from text descriptions of genes instead (Chen and Zou 2025). Genuine context-dependent gene function exists. Some of it acts at the protein level, as when a moonlighting protein changes function with its location or binding partners (Jeffery 1999), and need not change the gene’s expression. Some acts through transcription, as when lineage-determining transcription factors select different enhancers with different partners by cell type (Heinz et al. 2010).

Aim. We organise the question into two levels. Level 1 (a fact about the model): is a gene’s representation reshaped by context in a structured, gene-specific way, is that reshaping learned, and is it linked to what the model does? Level 2 (biology): does the reshaping reveal something about the gene’s role in each context beyond what co-expression already tells us? We study Level 1 for MaxToki-217M in depth (rank control, independent replication, random-weights control, link to the model’s predictions, steering) and repeat the central tests in MaxToki-1B, scGPT and STATE on identical cells. For Level 2, we test the functional organisation of the contextualisation against co-expression nulls and against representations built from expression alone, and run two direct tests for context-specific gene function, in all four models. The study makes three contributions. (1) It measures how much, and how gene-specifically, a gene’s representation changes between cell types (Level 1). It also tests whether the model reacts when functional directions are pushed into it, and whether genes that change more gain more from their own context in next-gene prediction. Neither test shows that the natural change is used. (2) It shows that this change is organised by co-expression, and that two tests find no context-specific gene function beyond co-expression in any of the four models (Level 2). (3) It tests whether the functional organisation of the change can be used to compare models, and where that comparison breaks down.

Methods

Models

MaxToki-217M and MaxToki-1B (Gómez Ortega et al. 2026; Theodoris Lab 2026) are Llama-architecture autoregressive models over rank-ordered gene tokens, with 11 and 20 transformer layers and hidden widths of 1232 and 2304. We used the Hugging Face safetensors checkpoints through the transformers library with `output_hidden_states=True`, in float32. For MaxToki, “layer k ” is the k -th hidden state: layer 0 is the embedding output and layer $k \geq 1$ is the output of the k -th transformer block; the last hidden state (layer 11 of MaxToki-217M) is taken after the model’s final normalisation. scGPT (Cui et al. 2024; Bo Wang Lab, n.d.) (whole-human checkpoint; 12 layers; width 512) and STATE (Adduri et al. 2026; Arc Institute 2025) (the SE-600M state-embedding model; 16 layers; width 2048) were used for the cross-architecture comparison. STATE’s gene tokens are fixed protein embeddings from the Evolutionary Scale Modeling 2 (ESM2) language model (Lin et al. 2023), passed through a learned projection together with each gene’s expression. For all models, “layer k ” is the representation after k transformer blocks, so models can be compared at matched relative depth (k divided by the number of blocks). As untrained controls, we built each architecture with random initial weights (the library’s default initialisation, seed 0): a LlamaForCausalLM from MaxToki-217M’s configuration with the same vocabulary, scGPT’s TransformerModel with the whole-human configuration, and STATE’s model with all weights random except its gene table. STATE never trains this table (fixed ESM2 protein embeddings), so it is the same in the trained and untrained models; the learned projection that reads it is random.

Datasets and annotations

The main data are the Tabula Sapiens (Tabula Sapiens Consortium 2022) immune, kidney and lung datasets from CZ CELLxGENE Discover (CZI Cell Science Program et al. 2025; Tabula Sapiens Consortium, n.d.-a, n.d.-b, n.d.-c). For the immune dataset we used a 20,000-cell subset; its cell identifiers are listed in the code repository. These files hold $\log(1 + \text{counts per } 10,000)$, written `log1p` below, in their main matrix, computed from counts corrected for ambient RNA, and the original integer counts in their raw matrix. All model inputs, the expression-only representations and the co-expression matrix were computed from the raw counts, so they include ambient RNA. Most selected cells come from 10x 3' assays; about 4% are Smart-seq2 cells, whose counts are read counts. Contexts are cell types, with cells pooled over the three datasets: the 12 cell types with the most cells (macrophage, pulmonary alveolar type 2 cell, kidney epithelial cell, capillary endothelial cell, CD4-positive alpha-beta T cell, CD8-positive alpha-beta T cell, B cell, basal cell, pulmonary alveolar type 1 cell, intermediate monocyte, neutrophil and classical monocyte), each with at least 1000 cells. We took 1000 randomly chosen cells per cell type (seed 0). Some cell types come from one dataset and others mix two or three (for example, most macrophages are lung cells), so a shift between two cell types can also reflect a different tissue mix.

For independent replication we used the human CD34+ bone-marrow data of Setty et al. (Setty et al. 2019) (5780 cells) as distributed with scVelo (Bergen et al. 2020; scVelo project, n.d.), which holds velocity spliced and unspliced integer counts for 14,319 pre-filtered genes (11,842 in the MaxToki vocabulary); its main matrix holds the spliced counts only. Because most Tabula Sapiens counts come from 10x assays that also count intronic reads, the main Setty analysis ranks genes by the sum of spliced and unspliced counts; a robustness run used spliced counts only. Contexts are the nine differentiation clusters with at least 130 cells: two haematopoietic stem cell clusters, precursors, common lymphoid progenitors, two erythroid clusters, two monocyte clusters and dendritic cells. Megakaryocytes (61 cells) were excluded, and all 5719 cells of the nine kept clusters were used.

Gene Ontology (GO) annotations (Ashburner et al. 2000; Gene Ontology Consortium et al. 2023) were taken from the gene-to-GO file distributed with GEARS (Roohani et al. 2024, 2022). A gene belongs to a functional pole only if it is annotated directly to one of the listed GO terms; annotations were not propagated through the GO hierarchy. Curated transcription-factor (TF) targets were all human TRRUST v2 interactions (Han et al. 2018b, 2018a) together with DoRothEA interactions of confidence A or B (Garcia-Alonso et al. 2019; Saez-Rodriguez Lab, n.d.). Gene symbols were mapped to Ensembl identifiers with the Geneformer dictionaries (version gc104M) (Theodoris et al. 2023; Theodoris, n.d.).

Input encoding and extraction of contextual gene representations

MaxToki uses Geneformer's rank-value encoding (Theodoris et al. 2023): a cell's counts are normalised to the cell total, each gene is divided by its median from Geneformer's gc104M gene-median dictionary (MaxToki's vocabulary holds the same 20,271 genes; the released MaxToki checkpoints include no median dictionary of their own), and the non-zero genes are ranked in decreasing order. A per-cell scale factor does not change the order within a cell, so the order is that of raw count divided by gene median. The first 1022 genes were kept and wrapped in the beginning- and end-of-sequence tokens (1024 tokens at most), a quarter of the model's 4096-token limit, to limit compute. Most cells express more genes than this, so MaxToki saw roughly the top 40% of a typical cell's genes, ranked by count divided by gene median. On real cells, this ordering gives the same token sequences as a separate rank-value tokenizer we wrote earlier for MaxToki, except that genes whose count-to-median values are equal after rounding to 32-bit floating point can appear in swapped order; we did not compare against an official MaxToki tokenizer.

We recorded, for each gene in a panel, its hidden state at that gene's token position and averaged it over the cells of one cell type. The main MaxToki-217M extraction used layers 0, 4, 8 and 11. A separate extraction of all 12 hidden states (the embedding output and all 11 transformer layers) with a 4000-gene panel, on the same cells, gave the full depth profile. Four design choices guard against known sources of false signal. (1) Pairs of cell types, not only genes found in every cell type: each per-gene mean is built from at most 50 occurrences in each half of the cells (see (3)), and we call a gene count-balanced in a cell type if it reaches that cap in both

halves. In the main extraction only 137 genes are count-balanced in all 12 cell types, many of them broadly expressed housekeeping genes (the single-cell analogue of the “stopwords” that are most context-sensitive in language models (Ethayarajh 2019)), whereas each pair of cell types shares a median of 1391 count-balanced genes. So we compare cell types two at a time. The panel contains genes with at least 25 occurrences in at least two cell types (the 6000 most frequent such genes). (2) Cell-level split halves: the cells of each cell type were split into two equal halves (500 cells each in the main extraction, 300 each in the 600-cell runs). So every per-gene mean is estimated twice, from different cells, and a result counts only if it repeats in both halves. (3) An occurrence cap: a per-gene mean has a standard error proportional to $1/\sqrt{n}$, and unequal counts inject context-dependent noise into any interaction term, so at most 50 occurrences were accumulated per (gene, cell type, partition), over a shuffled cell order. A (gene, cell type) entry is used only if it is count-balanced, that is, it reached the cap in both partitions. (4) Per-dimension z-scoring: before EXCESS, the functional axes and every other cosine-based measure, every hidden dimension was standardised over all count-balanced entries, to remove the inflation by a few dominant (“rogue”) dimensions that a scalar anisotropy correction misses (Timkey and van Schijndel 2021). The anisotropy check (below) also reports cosines on the raw representations, for comparison.

Other extractions used the same procedure with these settings. Scaling pair: MaxToki-217M and MaxToki-1B on the same 600 cells per cell type with a 5000-gene panel (cap 50), at layers 2 and 4 of MaxToki-217M and layers 4 and 7 of MaxToki-1B, so that the models can be compared at matched relative depth (layer 4 of 11 = 0.36 versus layer 7 of 20 = 0.35; layer 2 of 11 = 0.18 versus layer 4 of 20 = 0.20). Random-weights control: the same cells and the same 5000-gene panel as the MaxToki-217M scaling run, at layer 4. Setty data: cap 25, a floor of 15 occurrences in at least two clusters, a 5000-gene panel, layers 2 and 4.

scGPT and STATE were run on exactly the same cells, cell types and partitions as the 600-cell MaxToki runs, with the same 5000-gene panel and cap 50, each with its own standard input. scGPT (following its checkpoint configuration and its cell-embedding pipeline): the raw counts of all expressed genes in scGPT’s vocabulary were quantile-binned into 50 bins within the cell, 1199 of them were then sampled at random if the cell had more, and the `<cls>` token was prepended; layers 2, 4 and 9 (relative depth 0.17, 0.33 and 0.75). STATE (through its own data collator): raw counts over the genes in its protein table, the top 2047 expressed genes by count; we kept only positions holding expressed genes; layers 3, 6 and 12 (relative depth 0.19, 0.38 and 0.75). Of the 5000 panel genes, 4964 are in scGPT’s vocabulary and 4969 in STATE’s protein table. Each model’s own count-balanced entries were used. The untrained controls used the same cells and panel.

Expression-only representations

To test directly whether expression alone produces the functional organisation, we built contextual gene representations with no model, from the same cells, cell types and partitions as the 600-cell runs. Landmark version: in each cell type and partition, gene g was represented by the vector of Pearson correlations, across those 300 cells, between g ’s $\log_{10}(\text{counts per } 10,000)$ and that of each of 512 landmark genes (the most variable panel genes over all selected cells). Principal-component version: the same, with the scores of 256 principal components of the z-scored expression of all selected cells in place of the landmarks. Both are correlation profiles within one cell type, so they cannot see genes that rise and fall together between cell types. Our co-expression measure (next sections) pools all cell types and so includes that part. We therefore added a third version, built on axes from all cell types pooled, that we expected to respond to that part. Each of its vectors is still an average over cells of one cell type, divided by the gene’s total expression there, so the gene’s level in each cell type cancels out; it may therefore not capture co-expression between cell types either, and we did not check this directly. Centroid version: gene g in a cell type and partition was placed at the expression-weighted mean position of its expressing cells in the first 50 principal components of the same pooled, z-scored expression, that is, where in the pooled cell space the gene is expressed within that cell type. As in the model extractions, the cells of each cell type and partition were put in a random order and only the first 50 cells expressing the gene were used, so each vector averages as many cells as a model vector averages occurrences. Without this cap a vector averages all expressing cells (a median of 202 per entry; 10th to 90th percentile 108 to 277); we report that version only as a check. We first built the centroid version with 512 components and no cap. It failed the positive control: the shift shared by all genes did

not replicate between the two halves of the cells (cosine -0.22), because z-scoring each dimension gives the many noisy high-numbered components equal weight. With 50 components, a common default in single-cell analysis, it passes (replication $+0.79$); we did not run other numbers of components through the pipeline. The 512-component and all-cells results are in Additional file 1. The landmark and principal-component versions are correlations across all 300 cells of each half, so they are not matched to the models' 50 occurrences. All three used the same gene panel and the same count-balanced entries as MaxToki-217M and went through the identical pipeline (EXCESS, functional axes, all nulls). "Layer" does not apply to them.

Two-way decomposition and contextualisation strength (EXCESS)

For gene g in cell type c , we write $v(g, c) = \mu + a(g) + b(c) + E(g, c)$: a gene main effect $a(g)$ (the gene's identity), a context main effect $b(c)$ (a shift shared by all genes in that cell type, expected because attention pools over the cell), and a gene-specific interaction $E(g, c)$, the only term where a gene-specific response to context can appear.

For a cell-type pair (c_0, c_1) , the relative shift of gene g is $\delta(g) = [v(g, c_1) - v(g, c_0)] - \text{mean}_g[v(g', c_1) - v(g', c_0)]$: the gene's own shift minus the average shift of all genes. We computed δ separately in the two cell partitions (δ_0, δ_1) and defined $\text{EXCESS} = \text{mean} \cos(\delta_0(g), \delta_1(g)) - \text{mean} \cos(\delta_0(g), \delta_1(\text{perm } g))$: the same gene's shift agreeing across independent cells, minus a gene-shuffled control. $\text{EXCESS} = 0$ means no gene-specific response to context. The means are pooled over all (gene, pair) entries from the cell-type pairs with at least 200 count-balanced genes (150 in the cross-model comparison of contextualisation strength and in the Setty analysis), which equals a mean over pairs weighted by their number of genes. In the 600-cell data three of the 66 pairs have 150 to 199 such genes, so EXCESS in the cross-model comparison can differ by up to 0.003 from EXCESS in the functional-organisation controls. 95% confidence intervals (CIs) come from 2000 bootstrap resamples of genes, in which each resampled gene brings all of its pair entries. As a positive control we report the replication of the context main effect across partitions (the cosine between the two partitions' mean shifts).

Anisotropy. On the raw (unstandardised) representations we computed the mean cosine of random different-gene pairs (the anisotropy baseline (Ethayarajh 2019)), same-gene self-similarity across cell types, and the share of variance on the first principal component (PC), and repeated these after per-dimension z-scoring. Self-similarity was averaged over all genes count-balanced in at least two cell types (5433 at layer 4). The anisotropy baseline is the mean over 6000 random pairs of (gene, cell type) entries, and the first-PC share comes from 4000 randomly drawn entries.

Rank-position control. MaxToki encodes token position, and a gene's rank changes between cell types. We computed each gene's mean rank in each cell type over the cells in which it was among the 1022 genes given to MaxToki (only these cells give the gene a representation), using the same cells as the extraction. We split genes at the median absolute rank change into rank-stable and rank-moving halves and recomputed EXCESS in each, correlated the size of the shift with the rank change, and recomputed EXCESS after a linear regression of the shift on the gene's mean rank in each of the two cell types and on its absolute rank change. Because rank could act in a curved way, we repeated the regression with squared, product and log terms of the ranks. As a second check, we replaced the gene-shuffled control with a rank-matched one: each gene's shift was compared with that of the gene nearest to it in (rank in one cell type, rank in the other). If rank drove the shift, genes with the same ranks would share it.

Functional axes and reproducible interaction power

A functional axis u is the unit vector separating two GO-defined gene classes in the gene main effect, $u \propto \text{mean}(a(g) \mid \text{class A}) - \text{mean}(a(g) \mid \text{class B})$, where $a(g)$ is the gene's z-scored representation averaged over partitions and count-balanced cell types. Genes annotated to both poles were removed from both. The axes were nuclear versus surface/secreted (nucleus GO:0005634, chromatin GO:0000785 and DNA binding GO:0003677, versus plasma membrane GO:0005886, extracellular region GO:0005576 and extracellular space GO:0005615); mitochondrion (GO:0005739) versus cytoskeleton (GO:0005856); transcription (regulation of DNA-templated transcription GO:0006355 and DNA-binding transcription factor activity

GO:0003700) versus transport and signalling (ion transport GO:0006811 and signalling receptor activity GO:0038023); and cell cycle (GO:0007049) versus cell differentiation (GO:0030154). Axis validity is the five-fold cross-validated area under the receiver operating characteristic (ROC) curve (AUC) with which $a(g, u)$ separates held-out pole genes.

Reproducible interaction power measures how much context moves genes along u in a way that replicates across independent cells. For genes count-balanced in at least 9 cell types, we projected each partition's representation onto u , $q_p(g, c) = \langle v_p(g, c), u \rangle$; regressed out the gene's mean rank in that cell type (linear and quadratic terms; "rank-controlled"); took the two-way interaction $I_p(g, c)$ (the projection minus its gene mean and context mean, plus the grand mean); and averaged $I_0(g, c) \cdot I_1(g, c)$ over all (gene, cell type) entries. Noise averages to zero in this product, so only modulation that reproduces across the two partitions survives. Functional-z is the z-score of this power against 300 axes built from random gene partitions of the same pole sizes. We report it at layers 4, 8 and 11, without and with the rank regression. A cell type's loading on an axis is the mean projection onto u of the z-scored layer-4 representations (averaged over partitions) of the genes count-balanced in that cell type. These gene sets differ between cell types and the loadings are not rank-controlled, so we use them only to describe the ordering of cell types.

Null models for functional organisation

Functional gene sets may be co-expressed, and co-expressed genes co-vary across cells whatever the model "understands". We therefore compared each of the three axes with well-separated poles (nuclear/surface, mitochondrion/cytoskeleton and transcription/transport; the cell-cycle/differentiation axis separates its poles only weakly and was not tested) with null axes whose two poles have the same sizes as the functional poles, using the rank-controlled power. Null 1 (random axes) is the null behind functional-z and was computed at layers 4, 8 and 11; its empirical p was computed at layer 4. Nulls 2 and 3 were computed at layer 4, and null 4 at layers 4 and 8. In the 600-cell comparison each model was tested at its own depth-matched layer. Co-expression was the Pearson correlation of $\log_{10}(\text{counts per } 10,000)$, computed from the raw counts over 3000 cells from each of the three Tabula Sapiens datasets, and the coherence of a gene set was its mean pairwise correlation.

1. **Random axes:** random genes with the functional poles' sizes (300 axes). For functional-z the two poles do not overlap. For the empirical p at layer 4 we drew a separate set of 300 random axes, with the two poles chosen separately so that they can share genes.
2. **Strongly co-expressed modules:** each null pole was the set of genes most correlated with a random seed gene (300 axes). These modules are much more co-expressed than the functional poles (Results), so this is a strong null rather than a matched one.
3. **Size- and co-expression-matched sets:** each null pole was built from the k genes most correlated with a random seed gene plus randomly chosen genes, with k chosen by bisection so that the pole's coherence was within 10% of the functional pole's (300 axes). We also kept pole size fixed and traced power against the share of each pole drawn from the genes most co-expressed with the seed gene (0 to 100%, 25 draws per step).
4. **Size- and tightness-matched sets:** the same construction with representation tightness (the mean pairwise cosine of the pole genes' main effects $a(g)$) in place of co-expression (300 axes). When a functional pole was less tight than random genes, so that even a random set was at least as tight, a random set of the same size was used for that pole. This happened in at most 11 of 300 null axes for any axis in the four trained models. In the landmark expression-only version it happened in at most 36 of 300. In the centroid expression-only version it happened in at most 3 of 300. The other pole of each axis was always matched. Such sets make the null harder, not easier, to beat.

For each null, the empirical one-sided p is $(k + 1)/(N + 1)$, where k is the number of the $N = 300$ null axes with power at least as large as the functional axis; when no null axis reached it, p takes its smallest possible value, $1/301$. Empirical p values were used because the null power distributions are right-skewed.

Causal steering test

To test whether the model uses the functional direction, we added it to the residual stream at half of a cell's gene positions and read the model's own next-gene output logits at the other half. The direction was the raw (unstandardised) difference between the mean layer-4 representations of the nuclear-pole and surface-pole genes (poles made disjoint, as for the axes). It was added at the layer-4 representation (the output of the fourth transformer block). For each of 30 immune cells (at most 512 tokens), gene positions were randomly split into a steered half and a read half, so any effect at the read positions must travel through attention. The readout used a wider gene set than the direction: every gene token in MaxToki's vocabulary annotated to a pole (4844 nuclear and 5520 surface tokens; genes annotated to both removed). The readout (specificity) was the mean change in logit over the nuclear-pole tokens minus that over the surface-pole tokens. Strengths were 0.25, 0.5 and 1.0 times the mean residual-stream norm at that layer in one reference cell; we report 0.5 as the headline. Pushing toward $+u$ and toward $-u$ and taking the difference (the signed swing) cancels any effect that does not depend on the sign of the push. The first control was a random direction of the same norm, drawn fresh for each cell; significance was a one-sided sign test of functional swing greater than random swing across cells. This tests whether the push has a consistent effect in the expected direction. Because a random direction is not built like the axis, a second control was matched to the axis: 30 fixed random splits of the pooled nuclear-pole and surface-pole genes into two groups of the pole sizes, each turned into a direction exactly like the functional axis and applied to the same cells, positions and strength (0.5). The orientation of a random split is arbitrary, so we compared sizes: $p = (1 + \text{number of splits whose absolute mean swing is at least the absolute functional swing}) / (30 + 1)$, whose smallest possible value is $1/(30 + 1) = 0.032$. We read the direction from the sign of the functional swing. With 30 splits these p values cannot survive a correction for several tests, so we report them uncorrected, together with z , the number of standard deviations by which the functional swing exceeds the absolute split swings. The test was repeated at the layer-8 representation and for the mitochondrion/cytoskeleton and transcription/transport axes at layer 4. To put the push sizes in context, we measured how far context naturally moves genes along the same direction: the spread of each gene's projection onto u across its count-balanced cell types.

To compare MaxToki-217M with MaxToki-1B, we repeated the steering test on the 600-cell extractions at matched relative depth (MaxToki-217M layer 4, MaxToki-1B layer 7), with a different push unit. MaxToki-1B has two hidden dimensions that are very large in every gene's representation. Its raw residual norm is 3.0 times the norm of the part that differs between genes, against 1.1 times in MaxToki-217M, so a push scaled to the raw norm would be much stronger in MaxToki-1B. In this comparison the push unit was therefore the mean norm of the residual after removing its mean over the cell's gene positions (averaged over 5 cells). We ran all three axes in both sizes, with both controls, and repeated the nuclear/surface test in both sizes with the axis, and the split directions, set to zero in the two largest dimensions. The 1000-cell configurations above were also run against the fixed splits.

Tests for context-specific gene function

Directional congruence. Along each functional axis, we decomposed each gene's projection into a gene loading $f(g)$ (mean over contexts), a context loading $h(c)$ (mean over genes) and an interaction $e(g,c)$. If genes move toward their own functional pole specifically in cell types that share that character, e should follow $f \cdot h$. We estimated f and h in one partition and e in the other (and vice versa), measured $\beta = \text{corr}(e, f \cdot h)$ on rank-residualised projections, and compared β with 300 null axes whose poles were matched to the functional poles on size and co-expression (as in null 3 above).

Placement near curated targets. For TFs count-balanced in at least 6 cell types with at least 15 curated targets in the gene panel, we asked (A) whether, in each cell type, the TF's representation (with the context mean removed) is closer by cosine to the centroid of its curated targets than to the centroid of non-targets matched on co-expression with the TF (200 redraws), and (B) whether this excess closeness rises in cell types where the targets are more highly expressed (Spearman correlation across cell types with target activity, the negative of the targets' mean rank). The matched non-targets are matched on co-expression with the TF but not on gene type, and curated target lists are rich in regulators, so (C) we also asked whether the TF is closer

to its own targets than to the targets of the other TFs tested (same cell type, same measure). A cell type was used for a TF only if at least 8 of its targets were count-balanced there, and a TF needed at least 3 such cell types. All three questions were tested across TFs with sign tests.

Lineage transcription factors. We also tried a test on 14 TFs with lineage-specific roles in immune cells (SPI1, IRF4, IRF8, GATA3, MAFB, BCL6, PRDM1, TBX21, RUNX1, RUNX3, BATE, MAF, BACH2 and ZEB2) (Heinz et al. 2010). It asked whether these TFs move more between cell types than other genes do. A gene’s movement was the part of its gene-specific shift in each cell type that repeats in both cell halves (the product of the two halves’ shifts, averaged over cell types; MaxToki-217M, 1000-cell extraction, layers 4 and 8). The controls were genes matched on abundance, and genes matched on abundance and on how much their co-expression partners change between cell types; p values came from a permutation that keeps TF families together. Because movement is averaged over cell types, the test needs genes count-balanced in at least 9 of the 12 cell types (Results).

Cross-model, scaling and random-weights comparison

All 600-cell representations (MaxToki-217M, MaxToki-1B, scGPT, STATE, the three untrained models and the three expression-only representations) share the cells, cell types, partitions and gene panel. Model vectors and centroid vectors average 50 occurrences or expressing cells per entry; the landmark and principal-component versions use all 300 cells of each half. EXCESS (on context pairs with at least 150 count-balanced genes) and nuclear/surface functional-z were computed the same way for every representation. Each trained model was scored at two depths (MaxToki-217M layers 2 and 4; MaxToki-1B layers 4 and 7; scGPT layers 2 and 4; STATE layers 3 and 6) and each untrained model at its depth-matched layer. In this comparison functional-z was computed without the rank regression, on genes count-balanced in at least 6 of the 12 cell types, against 1000 random axes whose two poles do not overlap. In the Setty replication it used genes count-balanced in at least 4 of the 9 clusters and 200 random axes, whose two poles were drawn separately, so they could share genes. Each model has its own count-balanced entries, so we repeated the comparison using only the (cell type, gene) entries that are count-balanced in every representation. The rank regression, all four nulls and both Level 2 tests were also run on the four trained models at their depth-matched layers (MaxToki-217M layer 4, MaxToki-1B layer 7, scGPT layer 4, STATE layer 6) and on the three expression-only representations. For the untrained models we also ran the curated-target test, and for untrained scGPT and STATE the rank regression on the functional axes. On the untrained models only random axes (null 1) were used: we ran none of the co-expression and tightness nulls (nulls 2 to 4) and not the directional-congruence test on them. We built no untrained MaxToki-1B. With 600 cells per cell type fewer genes reach the cap in many cell types, so these runs used genes count-balanced in at least 6 cell types. For the functional axes and the co-expression nulls, at least 9 cell types gave the same conclusions (Additional file 1); the Level 2 tests were run only with at least 6. For the rank regression of non-MaxToki representations we used MaxToki’s mean rank on the same cells as an abundance covariate. For the functional axes and the two co-expression nulls we repeated it with the gene’s mean $\log_{10}(\text{counts per } 10,000)$ instead.

Link to next-gene prediction

For 350 Tabula Sapiens immune cells with at least 60 expressed vocabulary genes (at most 512 tokens), and up to 20 target positions per cell from the sixth gene onward, we compared the model’s log-probability of the true gene given the cell’s own preceding genes with that given the same number of preceding genes from another cell (a chimeric context). The donor was another cell, chosen at random, with at least as many genes as the real prefix. The donor genes used could not include the target gene, so the chimeric context never contains it. The per-gene context benefit (log-probability difference, averaged over occurrences; genes with at least 3 occurrences) was correlated with the gene’s context movement: the mean, over the cell types where the gene is count-balanced (at least 4), of $\|z(g,c) - \text{mean}_c z(g,c)\|$ in the z-scored layer-4 space. This is one number per gene, unlike EXCESS. We used a partial Spearman correlation that controls for the log of the number of times the gene was sampled as a target, with a 95% CI from 2000 bootstrap resamples of genes.

To compare MaxToki-217M and MaxToki-1B, we ran a paired version on the same 350 cells, with each gene’s

context movement taken from the 600-cell extractions (MaxToki-217M layer 4, MaxToki-1B layer 7). The cells, target positions and donor cells were drawn from seeds that do not depend on the model, so both models were scored on identical (cell, position, donor) triples. Because MaxToki is causal, the log-probabilities for every prefix of a cell come from one forward pass over the whole cell; on 20 random queries per model this matched separate prefix passes to within 1.2×10^{-5} nats (largest difference). A second donor arm drew the donor from the same cell type, which asks whether context helps beyond cell identity.

Statistics and reproducibility

All random draws use fixed seeds (base seed 0), so every script draws the same cells, genes and null sets each time it runs. Most analyses work from saved hidden states; the steering and prediction-link tests run the model again. Empirical p values are one-sided unless stated. We corrected for multiple tests in three families. Steering (1000-cell MaxToki-217M): at each strength separately, the random-direction sign tests of the four configurations (nuclear/surface at layers 4 and 8; mitochondrion/cytoskeleton and transcription/transport at layer 4), with Bonferroni; the three strengths are not corrected against each other. Curated targets (1000-cell MaxToki-217M): static closeness and its rise with target activity, at layers 4 and 8, with Bonferroni. Directional congruence: three axes at layers 4 and 8 in the 1000-cell extraction, with Holm, counting only positive β ; the same Holm correction was applied over each 600-cell representation’s own tests (Table 4). The fixed-split p values are not corrected (see Causal steering test). All other p values are not corrected, including the null p values (Table 3), the own-versus-other-targets test, the curated-target tests on the 600-cell representations (Table 4) and the 600-cell steering tests. Where Results gives a corrected value for one of these, it names the correction. Each analysis has its own script: input encoding `ctx_tokenise.py`; cell selection `ctx_cell_selection.py`; extraction `ctx_extract_maxtoki.py`, `ctx_extract_scgpt_std.py`, `ctx_extract_state_std.py`, `ctx_extract_devel.py`, `ctx_extract_random.py` and `ctx_extract_expression.py`; run configuration for any extraction `ctx_prefix.py`; shared code `ctx_coexpr_null.py` (only its co-expression matrix function is used; run on its own it gives an older null with the wrong pole sizes, which `ctx_coexpr_null_v2.py` and `ctx_coexpr_null_v3.py` replace), `steer_lib.py` (steering and model outputs) and `ctx_stats.py` (gene bootstrap); `ctx_polysemy.py`, `ctx_layer_curve.py` and `ctx_anisotropy.py` (contextualisation by layer and anisotropy); `ctx_position_confound.py` (rank control); `ctx_independent.py` (Setty replication); `ctx_functional_axes.py`, `ctx_celltype_loadings.py`, `ctx_coexpr_null_v2.py`, `ctx_coexpr_null_v3.py` and `ctx_tightness_null.py` (functional axes and nulls); `ctx_causal.py` (steering); `ctx_directional_probe.py`, `ctx_curated_targets.py` and `ctx_switcher_test.py` (tests for context-specific function); `ctx_cross_model.py` (cross-model comparison); `ctx_context_composition.py` (dataset, donor and assay of each cell type’s cells); `ctx_expression_cellcounts.py` (cells behind each expression-only vector); `ctx_prediction_link.py` (prediction link); and `ctx_figures.py` (figures). The cell-selection and extraction scripts save .npz files; the analysis scripts write `results/ctx_*.json`, from which every number and table is taken, and `ctx_figures.py` draws the figures from the same files. The runs used Python 3.12 with PyTorch 2.12.1, transformers 5.13.0 and NumPy 2.4 on Apple-silicon hardware.

Use of large language models

An artificial intelligence assistant (Claude, Anthropic) was used to help write and run the analysis code, check references, and draft and edit the manuscript text. The author designed the study, reviewed the code, results and text, and takes full responsibility for the content. The assistant is not an author.

Results

The first sections use the main MaxToki-217M extraction (1000 cells per cell type) unless a model is named; later sections compare all models on the same 600 cells per cell type. For Level 1, we show that contextualisation is real, is not explained by rank position and replicates on a second dataset, and that the nuclear/surface direction is read by the model when pushed, although not every functional direction is. Two later sections

show that contextualisation is partly learned and weakly linked to prediction. For Level 2, we compare the functional organisation with co-expression nulls and with representations built from expression alone, and run two direct tests for context-specific gene function.

Genes are contextualised gene-specifically; in MaxToki-217M most reproducibly in the first layers

The gene-specific context response is large and highly reproducible (Table 1, Fig. 1). At layer 4 of the main extraction, EXCESS is +0.758: the same gene’s relative shift agrees at cosine +0.759 across independent cells, while a different gene’s agrees at +0.001, so the effect is gene-specific. The context main effect replicates at +0.996, which shows that the measurement works. Layer 0 (the context-free embedding output) gives exactly 0.000, as expected: a gene has the same embedding in every cell, so it has no shift to measure. The control that tests the statistic itself is the different-gene agreement (+0.001). The full depth scan shows where the response arises: it is highest at layer 1 (+0.847) and layer 2 (+0.835) and then falls to +0.651 at layer 11 (Fig. 1). The gene-specific shift is therefore already present after the first transformer layer and is most reproducible at layers 1 and 2. EXCESS is a cosine, so it measures how reproducible the direction of the shift is, not how far a gene moves. Deeper layers keep most of it (90% of the peak at layer 4 and 84% at layer 8), while the gene-specific shift becomes somewhat less reproducible toward the output.

Table 1: **Table 1.** Gene-specific context response at the four layers of the main MaxToki-217M extraction

Layer	EXCESS (same – different)	Same gene	Different gene	Main-effect replication
0 (embedding)	+0.000	+0.000	+0.000	—
4	+0.758	+0.759	+0.001	+0.996
8	+0.711	+0.711	+0.000	+0.996
11	+0.649	+0.650	+0.001	+0.997

EXCESS is the agreement of the same gene’s relative context shift across two independent halves of the cells, minus the agreement for a different gene, pooled over 66 cell-type pairs (a median of 1391 count-balanced genes per pair). Gene-level bootstrap 95% CIs are within ± 0.003 . The full depth profile is shown in Fig. 1.

Metric validity (anisotropy). These representations are anisotropic, as expected (Ethayarajh 2019): the mean cosine of random different-gene pairs is +0.20 to +0.62 at layers 4–11 (raw). After per-dimension z-scoring, which brings anisotropy to about 0.00, same-gene self-similarity across cell types falls with depth: +0.88, +0.65 and +0.61 at layers 4, 8 and 11 (raw, uncorrected: +0.92, +0.85 and +0.69). So genes move more between cell types in upper layers, as Ethayarajh found in language models. Simply subtracting the raw anisotropy baseline gives +0.62, +0.24 and +0.48 instead. This series dips at layer 8, where raw anisotropy peaks, so the layer-8 value mostly reflects the baseline. We z-score each dimension before every cosine-based measure (Methods), because a few dominant dimensions can drive raw cosines and one subtraction does not remove them (Timkey and van Schijndel 2021). With all three versions, self-similarity is higher at layer 4 than at layer 11. Together with the EXCESS profile, this means that upper layers move genes more but less reproducibly. After z-scoring, the first principal component carries only 0.07 of the variance at layer 11, so no single direction dominates the cosines.

The effect is not explained by the gene’s rank position

MaxToki reads a cell as a rank-ordered gene list and encodes token position, so a gene ranking 5th in one cell type and 500th in another is an obvious confound. Three checks show that rank does not drive the gene-specific shift (Fig. 2). At layer 4 the response is about the same for genes whose rank changes less than the median (+0.749; median change 27 positions) and more than the median (+0.767; 107 positions). 99% of it survives a linear regression on the gene’s rank in each cell type and its rank change (+0.749), and 99%

survives a regression that also allows curved effects of rank (+0.747); at layers 8 and 11, 98% and 97% survive the linear regression. And a gene’s shift agrees only a little with that of the gene closest to it in rank in both cell types (+0.058, against +0.759 for the same gene and +0.001 for a random gene). Measured against this rank-matched gene instead of a random one, EXCESS is still +0.701 at layer 4 (93% of the full value), and 90% and 88% at layers 8 and 11. So genes with the same ranks share only a small part of the shift. This part is an upper bound on what rank explains, because genes with the same ranks also tend to have similar expression levels. The size of a gene’s shift correlates only weakly with its rank change ($\rho = 0.26$ at layer 4 and 0.31 at layer 11). These checks cannot exclude every way in which position could enter a causal model.

The effect replicates on an independent dataset

In MaxToki-217M, on the Setty CD34+ bone-marrow data, with nine differentiation clusters rather than mature cell types as contexts, the phenomenon replicates: EXCESS is +0.713 at layer 2 and +0.713 at layer 4 (gene-shuffled null +0.0036 and +0.0039), and functional-z (without the rank regression, against random axes only; Methods) is +5.0 and +5.0 for the nuclear/surface axis and +16.1 and +8.8 for the mitochondrion/cytoskeleton axis. We did not run the rank control, the co-expression nulls or the Level 2 tests on these data. Using spliced counts only gives the same picture (EXCESS +0.702 and nuclear/surface functional-z +9.3 at layer 4). Neither the contextualisation nor its organisation along these two axes is specific to *Tabula Sapiens* or to mature cell types.

Context moves genes along functional axes, but not significantly more than along gene sets matched on co-expression

Context moves genes along interpretable functional axes far more than along random directions (Table 2). At layer 4, functional-z (rank-controlled in parentheses) is +16.6 (+13.3) for nuclear versus surface/secreted genes, +8.7 (+6.4) for mitochondrion versus cytoskeleton and +2.9 (+4.3) for transcription versus transport. The cell-cycle versus differentiation axis, which separates its poles only weakly (validity AUC 0.57), gives +2.0 (+0.9). The organisation persists in deeper layers for the two strongest axes. Along the nuclear/surface axis, lymphocytes sit at the nuclear end (CD8 T +4.7, CD4 T +4.4, B +3.9) and alveolar epithelial cells and macrophages at the surface end (alveolar type 1 -3.6, type 2 -3.2, macrophage -2.8). Each cell type is scored on its own set of expressed genes, so this order may mostly show which genes each cell type expresses, not how context moves the same genes (Methods).

Table 2: **Table 2.** Functional organisation of context modulation, by axis and layer (MaxToki-217M)

Axis	Poles (genes)	Validity AUC (layer 4)	Layer 4	Layer 8	Layer 11
Nuclear / surface	1800/1188	0.77	+16.6 (+13.3)	+7.8 (+6.5)	+6.1 (+6.0)
Mitochondrion / cytoskeleton	675/165	0.78	+8.7 (+6.4)	+12.7 (+8.4)	+10.4 (+6.9)
Transcription / transport	326/89	0.69	+2.9 (+4.3)	+1.0 (+1.3)	+0.8 (+0.8)
Cell cycle / differentiation	145/167	0.57	+2.0 (+0.9)	+2.9 (+1.7)	+2.6 (+1.9)

Functional-z is the z-score of the axis’s reproducible interaction power against 300 axes built from random gene sets of the same pole sizes, without (and, in parentheses, with) regression on the gene’s rank. Validity AUC is the cross-validated area under the ROC curve with which the axis separates held-out pole genes.

The key control asks whether this organisation exceeds co-expression (Fig. 3). Functional gene sets can be co-regulated, and co-regulated genes co-vary across cells whatever the model “understands”. The functional

poles are more co-expressed than random genes, but far less than co-expression modules (mean pairwise correlation 0.028 and 0.042 for the nuclear and surface poles, against 0.025 for random gene sets). At layer 4, against null gene sets matched on size and this average co-expression, no functional axis exceeds the null: the nuclear/surface power (0.50) sits inside the matched null (mean 0.40, 95th percentile 0.68; $p = 0.316$), and $p = 0.375$ and 0.083 for the other two axes. Modules built from the genes most co-expressed with a seed gene, 3–11 times more co-expressed than the functional poles, reach higher power on average (nuclear/surface null mean 0.67; $p = 0.701$, 0.757 and 0.804). We also kept pole size fixed and changed what share of each pole came from the genes most co-expressed with a seed gene. Null power rose steadily with that share. None of the 50 null axes with 0–10% such genes reached the nuclear/surface power (0.50), but 22 of 25 axes made only of such genes did. No single step matches both functional poles. The nuclear pole’s co-expression (0.028) is close to that of the 10% step (0.027), where 0 of 25 null axes reached the functional power; the surface pole’s (0.042) is close to that of the 35% step (0.043), where 10 of 25 did. The matched null, which matches each pole separately, puts the axis inside the null ($p = 0.316$). At layer 4 the organisation is not a by-product of compact gene sets in representation space: it exceeds sets matched on size and representation tightness ($p = 0.007$, 0.023 and 0.013). At layer 8 it does not ($p = 0.233$, 0.060 and 0.100). It far exceeds random axes (none of 300 reached it; $p = 1/301$ for nuclear/surface). So, at layer 4, the functional organisation is not significantly stronger than that of gene sets with the same size and average co-expression. This test can detect only large differences: all three axes sit above the matched-null mean, and an axis would need about 1.7–1.8 times the null mean to reach $p < 0.05$. We ran the co-expression nulls at layer 4 only, not at layers 8 and 11.

Expression alone gives a gene-specific change, but not its organisation

The nulls above ask whether functional gene sets are special among gene sets with the same co-expression, inside the model. A different question is whether a representation built only from expression shows the same contextualisation and organisation. On the same 600 cells per cell type, we compared MaxToki-217M (layer 4) with three expression-only representations (Methods). The landmark and principal-component versions use correlations within one cell type; the centroid version places each gene where it is expressed in a cell space built from all cells pooled. All three change with cell type in a gene-specific way (EXCESS +0.427, +0.218 and +0.321, against +0.738 for MaxToki-217M on the same cells), and 99% of the centroid version’s change survives the rank regression. So a reproducible, gene-specific change between cell types does not need a model: part of it follows from where each gene is expressed within each cell type. EXCESS depends on how many cells each vector averages. With all expressing cells instead of 50, the centroid version reaches +0.605, and the two correlation versions use all 300 cells of each half. These values should therefore not be ranked against the models’ (Methods). But none of the three is clearly organised along the functional axes. Rank-controlled functional-z is -1.4 , $+1.8$ and $+1.4$ (landmarks), $+0.6$, $+0.4$ and $+0.8$ (principal components) and -0.3 , -2.1 and $+0.5$ (centroids), against $+16.7$, $+7.5$ and $+5.0$ in MaxToki-217M (Table 3).

The difference also shows in how co-expressed gene sets behave. We express each null’s power as a multiple of the power of random poles in the same representation. In MaxToki-217M, sets matched on co-expression reach 1.8–6.0 times random, modules of strongly co-expressed genes 5.6–10.1 times, and poles made only of genes co-expressed with a seed gene 6.1–9.8 times. The other models behave the same way: strong modules reach 5.0–8.9 times random in MaxToki-1B, 3.3–4.2 times in scGPT and 12.5–18.0 times in STATE. The expression-only versions do not. The two within-cell-type versions barely respond to co-expression. Matched sets reach 1.4–1.6 times random with landmarks and 1.3–1.4 times with principal components, and strong modules about the same (1.4–1.5 and 1.5 times). In the centroid version, co-expressed sets do no better than random sets: strong modules reach 0.9–1.0 times random, and poles made only of co-expressed genes 0.9 times. So in every model, context moves genes much more along directions that separate co-expressed gene sets than along random directions, and in none of the three expression-only versions does it. This holds for co-expressed sets in general, not for functional sets in particular. It does not show that the models add information beyond co-expression: other expression-only constructions might come closer (Limitations).

The nuclear/surface direction is read by the model, but not every functional direction is

In the steering test (Fig. 4), adding the nuclear/surface direction to half of a cell's genes moves the model's own logits at the untouched half with the sign of the push. Nuclear-gene logits rise relative to surface-gene logits for a push toward the nuclear pole (specificity +0.085) and fall relative to them for a push toward the surface pole (-0.097), a signed swing of +0.182. A random direction of the same norm, drawn fresh for each cell, gives +0.001 on average, near zero by design. The swing is +0.079, +0.182, +0.184 at strengths 0.25, 0.5 and 1.0, so it stops growing above 0.5. At 0.5 it exceeds the random swing in 22 of 30 cells ($p = 0.0081$; Bonferroni-adjusted 0.032); at 1.0 in 20 of 30 cells ($p = 0.049$; Bonferroni-adjusted 0.20). The effect is stronger at the layer-8 representation (swing +0.232, 30 of 30 cells, $p = 9.3 \times 10^{-10}$). Against 30 fixed random splits of the same pole genes, built like the axis, the layer-4 swing is larger than 30 of 30 of them (0 of 30 at least as large; $p = 1/31$, $z = +2.6$) and the layer-8 swing larger than 30 of 30 (0 of 30; $p = 1/31$, $z = +6.1$). The direction therefore propagates through attention and changes the model's predictions, more than directions built from random groups of the same genes.

This does not generalise to every functional axis. The mitochondrion/cytoskeleton direction has no significant effect at strengths 0.25 and 0.5 (swing +0.024 and +0.067; above the random swing in 17 of 30 and 15 of 30 cells, $p = 0.29$ and 0.57). At strength 1.0 its swing is +0.269, above the random swing in 20 of 30 cells ($p = 0.049$, Bonferroni-adjusted 0.20). The split control was run only at strength 0.5. There the swing is not distinct from the random splits (18 of 30 splits at least as large; $p = 0.613$). The strength-1.0 swing was not compared with the splits. The transcription/transport direction has a reversed effect that grows with strength (swing -0.139, -0.268, -0.678). At strength 0.5 the functional swing exceeds the random one in only 2 of 30 cells (two-sided sign test $p = 8.7 \times 10^{-7}$; 2 of 30 at 0.25 and 0 of 30 at 1.0). At 0.5 its size also exceeds that of 30 of 30 random splits, but in the reverse direction. We have no explanation for the reversal; it shows that an axis that separates genes is not automatically read in the expected direction. The pushes are also large compared with natural variation. Across cell types, a gene's projection on the nuclear/surface direction has a median range of 2.62 (95th percentile 4.57). The middle push (7.6 units) is about 3 times that median range, and the largest (15.2 units) about 6 times. The smallest push (3.8 units) is about 1.4 times the median range, below the 95th-percentile range. It already gives a swing of +0.079, above the random direction in 23 of 30 cells ($p = 0.0026$; Bonferroni-adjusted 0.010); it was not tested against the random splits. Even this push moves half of a cell's genes along the axis at once, which natural context does not do. The test therefore shows that the model can read this axis, not that the natural, context-driven changes along it are used.

The nuclear/surface result holds in MaxToki-1B. On the 600-cell extractions at matched relative depth, with the push scaled to the part of the residual that differs between genes, the swing at strength 0.5 is +0.212 in MaxToki-217M and +0.205 in MaxToki-1B. It exceeds a random direction in 25 of 30 and 30 of 30 cells ($p = 1.6 \times 10^{-4}$ and 9.3×10^{-10}). Against 30 random splits, 0 of 30 and 0 of 30 are at least as large (mean absolute split swing 0.067 and 0.047; $p = 1/31$ and $1/31$, $z = +3.6$ and $+4.2$). In MaxToki-1B, 52% of the axis lies in the two very large hidden dimensions. With those coordinates set to zero, the swing is +0.324 (0 of 30 splits at least as large; $z = +6.3$), so they do not carry the effect. In MaxToki-217M, which has no such very large dimensions, only 13% of the axis lies in its own two largest dimensions. Even so, with them set to zero the swing falls from +0.212 to +0.131 (splits at least as large: 1 of 30; $p = 0.065$, $z = +1.9$), so the swing is no longer clearly above the splits. In this model part of the effect goes through its two largest dimensions. The other two axes behave differently in the two sizes. In MaxToki-1B both are read in the expected direction: the swing is +0.400 for mitochondrion/cytoskeleton ($z = +4.7$ against the splits) and +0.111 for transcription/transport ($z = +4.6$). In MaxToki-217M the transcription/transport direction is again read in reverse (-0.242), and the mitochondrion/cytoskeleton swing (+0.168) is not distinct from the random splits (7 of 30 at least as large; $p = 0.258$). So only the nuclear/surface direction is read as expected in both sizes and both extractions; in the larger model all three directions are.

No context-specific gene function beyond co-expression

Two direct tests for context-appropriate gene function find no context-specific signal beyond co-expression. (1) Directional congruence asks whether genes move toward their own functional pole in the cell types that

lean toward that pole (β , Methods). For no axis was β significantly larger than for null axes matched on size and co-expression. At layer 4, β is +0.16, +0.10 and +0.23 for the three axes, all positive but within the upper tail of size- and co-expression-matched null axes, whose means are near zero ($p = 0.116, 0.086$ and 0.053 ; none significant after Holm correction over 6 tests; at layer 8, β is $-0.16, -0.11$ and $+0.09$). So at layer 4 the trend points the expected way but is not significant; the null spread of β (standard deviation $0.08\text{--}0.13$) means the test could detect only clear congruence. (2) Placement near curated targets: across 26 TFs, at layer 4 a TF's representation is slightly closer to the centroid of its curated targets than to that of co-expression-matched non-targets (mean cosine excess +0.021; 20 of 26 TFs positive; sign test $p = 0.005$, Bonferroni-adjusted over 4 tests 0.019). Compared with the targets of the other TFs tested, a TF is only slightly closer to its own targets (+0.008; 17 of 26 TFs positive; $p = 0.084$), so the closeness is not shown to be specific to each TF's own targets; much of it may reflect where regulators and their usual targets sit in general. The closeness does not rise significantly in the cell types where the targets are more active (mean $p = +0.084$; 15 of 26 TFs positive; $p = 0.28$, Bonferroni-adjusted 1.00); the trend is positive but weak, and each TF's p rests on as few as 3 cell types. At layer 8 neither effect is significant (static $p = 0.423$; modulation $p = +0.097$, $p = 0.28$). We did not run an untrained model on these cells. On the 600-cell extraction only 11 TFs qualify (Table 4). There the closeness is smaller and not significant (+0.011; 8 of 11 TFs positive; $p = 0.113$). The untrained MaxToki-217M gives a mean no smaller (+0.030), but only 6 of 11 TFs are positive ($p = 0.500$). That 11-TF test has little power, so we cannot say whether the closeness is learned, or whether it holds beyond these 26 TFs. A third attempt, on 14 lineage TFs, is uninformative rather than negative. The test needs a gene to be count-balanced in at least 9 of the 12 cell types. 10 of the 14 TFs are in the 6000-gene panel, but only 1 (RUNX1) met that bar. These regulators are weakly expressed, or are expressed in only a few of the cell types we used.

MaxToki-1B, scGPT and STATE also contextualise genes, but show no context-specific gene function beyond co-expression

We ran the full set of controls and both Level 2 tests on every trained model and expression-only representation of the same 600 cells per cell type (Table 3, Table 4). Three results hold in every model ((1), (3) and (4) below), one holds only in some models ((2)), and one holds only in MaxToki-217M ((5)). (1) Every model contextualises genes strongly and gene-specifically (EXCESS +0.738, +0.764, +0.813 and +0.867 for MaxToki-217M, MaxToki-1B, scGPT and STATE), and almost all of it survives the rank regression (99%, 99%, 98% and 99%). (2) Organisation beyond random axes holds only partly. The MaxToki models organise the change along all three functional axes beyond random axes (rank-controlled functional- z for the nuclear/surface axis +16.7 and +8.0), whichever abundance covariate is used (+16.8 and +7.8 with mean expression). STATE does so too (+5.5, +9.8 and +4.0). But when the gene's mean expression is the covariate, STATE's nuclear/surface organisation disappears (+0.2). Its other two axes remain (+7.0 and +3.3). STATE and scGPT receive each gene's expression level as input; MaxToki receives only the genes' rank order. scGPT organises the change only weakly (+1.4, +2.3 and +3.5). (3) In no representation does any functional axis significantly exceed gene sets matched on size and co-expression (smallest $p = 0.07, 0.19, 0.23$ and 0.32), or modules of strongly co-expressed genes. (4) No model shows context-specific gene function. Directional congruence is not significant after correction in any of the four models, and in none does a TF's closeness to its curated targets rise in the cell types where those targets are active. Expression alone can pass, or nearly pass, each test. In the centroid version we use, the closeness rises with target activity (mean $p = +0.214$; 9 of 11 TFs positive; $p = 0.03$, not corrected; 0.10 after Bonferroni over the three curated-target questions). Its build with all expressing cells, which we report only as a check, passes the directional test on the transcription/transport axis ($\beta = +0.11$, $p = 0.007$, significant after Holm correction over 3 tests), where its matched null is narrow (standard deviation 0.05 , against $0.11\text{--}0.16$ in MaxToki-217M); the build we use does not ($p = 0.113$). So a pass on either test would not by itself show gene function beyond expression. One static signal appears: STATE places TFs closer to their curated targets than to co-expression-matched non-targets (24 TFs, mean cosine excess +0.099, $p = 7.7 \times 10^{-4}$). Compared with the targets of the other TFs tested, a STATE TF is slightly closer to its own targets (+0.047; 17 of 24 TFs positive; $p = 0.032$, not corrected; 0.096 after Bonferroni over the three curated-target questions). So we cannot say that the closeness is specific to each TF's own targets. The untrained STATE, which uses the same fixed protein-embedding gene table as the trained model, shows

the same closeness ($+0.132$, $p = 1.4 \times 10^{-4}$) and is about as specific to each TF’s own targets ($+0.040$; 17 of 24 TFs positive; $p = 0.032$). Untrained scGPT and MaxToki-217M show no significant closeness (-0.008 , $p = 0.895$; $+0.030$, with 6 of 11 TFs positive, $p = 0.500$; this 11-TF test has little power). So STATE’s closeness does not come from training on expression. It most likely comes from the protein-sequence embeddings; we did not run STATE with a random gene table, which would test this directly. scGPT shows a different pattern. Its TFs are not closer to their targets than to co-expression-matched non-targets ($+0.026$; $p = 0.402$). But they are closer to their own targets than to the targets of the other TFs tested ($+0.070$; 13 of 16 TFs positive; $p = 0.011$, not corrected; $p = 0.011$ at layer 2), and untrained scGPT is not (-0.005 ; $p = 0.962$). This comparison is not matched on co-expression, so the scGPT result most likely reflects co-expression between a TF and its targets, learned in training, rather than a signal beyond co-expression. (5) Only MaxToki-217M passes the tightness control. It clearly exceeds gene sets matched on size and representation tightness on two of its three axes ($p = 1/301$ for nuclear/surface and $1/301$ for transcription/transport; mitochondrion/cytoskeleton $p = 0.053$). MaxToki-1B reaches $p = 0.033$ on one axis (transcription/transport); this p is not corrected and does not survive a Holm correction over the model’s three axes (0.10). scGPT and STATE exceed the null on no axis (Table 3). So only in MaxToki-217M is the organisation clearly not a by-product of compact gene sets.

Table 3: **Table 3.** Functional organisation and its controls in every representation, on the same cells

Representation	EXCESS	Kept after rank regression	Functional-z, rank-controlled	p, random axes	p, matched co-expression	p, strong co-expression modules	p, matched tightness
MaxToki-217M (layer 4)	+0.738	99%	+16.7 / +7.5 / +5.0	1/301 / 1/301 / 0.007	0.35 / 0.46 / 0.07	0.78 / 0.73 / 0.75	1/301 / 0.053 / 1/301
MaxToki-1B (layer 7)	+0.764	99%	+8.0 / +8.1 / +3.1	1/301 / 1/301 / 0.017	0.75 / 0.39 / 0.19	0.95 / 0.61 / 0.80	0.385 / 0.824 / 0.033
scGPT (layer 4)	+0.813	98%	+1.4 / +2.3 / +3.5	0.146 / 0.037 / 0.017	0.91 / 0.53 / 0.23	0.96 / 0.72 / 0.45	0.674 / 0.316 / 0.146
STATE (layer 6)	+0.867	99%	+5.5 / +9.8 / +4.0	0.010 / 1/301 / 0.017	0.71 / 0.36 / 0.32	0.95 / 0.64 / 0.82	0.369 / 0.143 / 0.289
Expression only, landmarks	+0.427	96%	-1.4 / +1.8 / +1.4	0.934 / 0.027 / 0.100	0.98 / 0.29 / 0.39	0.99 / 0.37 / 0.42	0.997 / 0.352 / 0.389
Expression only, principal components	+0.218	96%	+0.6 / +0.4 / +0.8	0.163 / 0.269 / 0.269	0.42 / 0.57 / 0.50	0.62 / 0.73 / 0.64	0.741 / 0.837 / 0.605
Expression only, centroids	+0.321	99%	-0.3 / -2.1 / +0.5	0.528 / 0.993 / 0.312	0.33 / 1.00 / 0.23	0.42 / 0.98 / 0.22	0.654 / 0.990 / 0.365

Each cell lists the nuclear/surface, mitochondrion/cytoskeleton and transcription/transport axes, in that order. All representations use the same 600 cells per cell type, partitions and gene panel; genes count-balanced in at least 6 of the 12 cell types. The p values are empirical one-sided values against 300 null axes of the same pole sizes (1/301 is the smallest possible). Rank regression uses MaxToki’s mean rank on the same cells as

the abundance covariate. Additional file 1 repeats the functional axes and the co-expression nulls with the mean log-normalised expression as covariate, and with genes count-balanced in at least 9 cell types.

Table 4: **Table 4.** Tests for context-specific gene function in every representation

Representation	Directional congruence β	p (smallest at this layer; Holm-significant tests, all layers)	TFs tested	Closeness to curated targets (p)	Closer to own than to other TFs' targets (p)	Rise where targets are active, ρ (p)
MaxToki-217M (layer 4)	+0.12 / -0.07 / +0.15	0.086 (0)	11	+0.011 (0.113)	+0.004 (0.887)	+0.028 (0.89)
MaxToki-1B (layer 7)	+0.07 / -0.20 / +0.02	0.306 (0)	11	+0.014 (0.274)	+0.001 (0.726)	+0.068 (0.50)
scGPT (layer 4)	-0.05 / -0.02 / -0.16	0.482 (0)	16	+0.026 (0.402)	+0.070 (0.011)	-0.002 (0.40)
STATE (layer 6)	-0.03 / +0.09 / -0.08	0.266 (0)	24	+0.099 (7.7 $\times 10^{-4}$)	+0.047 (0.032)	-0.047 (0.73)
Expression only, landmarks	-0.03 / -0.37 / +0.43	0.106 (0)	11	-0.098 (0.887)	-0.007 (0.726)	+0.037 (0.73)
Expression only, principal components	+0.04 / +0.14 / +0.25	0.206 (0)	11	-0.029 (0.887)	+0.014 (0.500)	+0.131 (0.50)
Expression only, centroids	+0.00 / -0.11 / +0.05	0.113 (0)	11	+0.049 (0.500)	+0.076 (0.500)	+0.214 (0.03)

Directional congruence β is listed for the nuclear/surface, mitochondrion/cytoskeleton and transcription/transport axes, against 300 size- and co-expression-matched null axes; the Holm correction runs over all axes and layers tested for that representation (6 tests for the MaxToki models, 9 for scGPT, 9 for STATE and 3 for the expression-only representations). The curated-target tests use TFs count-balanced in at least 6 cell types with at least 15 curated targets in the panel; with 600 cells per cell type fewer TFs qualify than in the 1000-cell MaxToki-217M extraction (26 TFs): 11 for the MaxToki models and the expression-only representations, 16 for scGPT and 24 for STATE. The 11-TF tests have little power. The third test compares a TF's closeness to its own targets with its closeness to the targets of the other TFs tested. The curated-target p values are not corrected for multiple tests.

Contextualisation across architectures and sizes, and with untrained models

Functional organisation can be measured whether or not a model shows context-specific gene function, so it could serve as a way to compare models (Table 5, Fig. 5). Here every representation shares the cells, cell types, partitions and gene panel. (1) Every model contextualises genes strongly (EXCESS +0.740, +0.767, +0.813 and +0.867 for MaxToki-217M, MaxToki-1B, scGPT and STATE at depth-matched layers; these differ slightly from Table 3 because more cell-type pairs are scored here, Methods), but contextualisation does not grow consistently with model size. At matched relative depth, MaxToki-1B gives +0.767 (layer 7) against

+0.740 for MaxToki-217M (layer 4), and +0.766 (layer 4) against +0.832 (layer 2). Functional organisation is not consistently higher in the larger model either: functional-z is +7.2 against +15.5 at relative depth 0.35, and +18.0 against +18.4 at 0.2. The depth pattern also differs between models. From the earlier to the later layer, EXCESS falls in MaxToki-217M (+0.829 to +0.738) and slightly in STATE (+0.879, +0.867 and +0.842 at layers 3, 6 and 12), stays the same in MaxToki-1B (+0.764 and +0.764) and rises in scGPT (+0.687, +0.813 and +0.872 at layers 2, 4 and 9; all depth values in this sentence use cell-type pairs with at least 200 count-balanced genes, as in Table 3). (2) On the nuclear/surface axis and each model’s own entries, functional organisation is highest in the MaxToki models and much lower in scGPT and STATE (+3.7 and +4.0). The order between the other models depends on the gene set and the controls. On the entries shared by all representations, STATE (+6.8) is above scGPT (+3.3) and MaxToki-1B at layer 7 (+4.8). After the rank regression, STATE is above both MaxToki models on the mitochondrion/cytoskeleton axis (+9.8 against +7.5 and +8.1; Table 3), but with the gene’s mean expression as the covariate it is below both (+7.0 against +9.1 and +8.8). On the nuclear/surface axis, only MaxToki-217M stays clearly above scGPT and STATE in every version: each model’s own entries, the shared entries, the 9-cell-type cut and both abundance covariates. On the other two axes it is not clearly above STATE. The models differ in more than input (training data, objective and gene vocabulary), so this comparison does not isolate the cause, and the functional organisation is not yet a stable way to rank architectures. (3) Part of the contextualisation needs no training in every architecture. Untrained versions keep 60% (MaxToki-217M), 36% (scGPT) and 88% (STATE) of the trained EXCESS. Untrained MaxToki-217M keeps a much smaller share of the functional organisation (functional-z +4.6 against +15.5; +4.1 against +11.3 on the shared entries). In scGPT and STATE the trained values are already low (+3.7 and +4.0), so these comparisons tell us little about whether training adds organisation. Untrained scGPT gives +2.2. Untrained STATE gives +1.4 on its own entries, but +3.8 against +6.8 on the shared entries. After the rank regression, untrained STATE keeps about half of the trained organisation on the nuclear/surface and mitochondrion/cytoskeleton axes (+3.0 against +5.5; +4.8 against +9.8) and is above the trained model on transcription/transport (+5.6 against +4.0). So in STATE much of the functional organisation does not need training. In untrained scGPT the functional axes barely separate their two poles (validity AUC 0.51–0.57), so its values cannot be read. We did not test why the untrained STATE keeps most of the contextualisation. Two things could contribute: each gene’s input token carries its expression level, which itself differs between cell types, and the untrained model uses the same fixed protein-embedding gene table as the trained one. Running STATE with random weights and a random gene table would separate the two; we did not run it. The expression-only representations show little or no functional organisation (−1.3, +0.7 and −0.4). Their EXCESS is lower (+0.424, +0.218 and +0.320), but it depends on how many cells each vector averages (+0.607 for the centroid version with all expressing cells), so the strength of contextualisation alone does not separate the models from expression. On the 14,109 shared entries EXCESS changes little. Besides the changes given above, on the shared entries MaxToki-1B is above MaxToki-217M at relative depth 0.2 (+15.1 against +13.5), and untrained STATE is above trained scGPT (+3.8 against +3.3).

Table 5: **Table 5.** Contextualisation strength and functional organisation of every representation on the same cells

Representation	Layer (relative depth)	Width	Genes	EXCESS	Functional- z	EXCESS, shared entries	Functional- z, shared entries
MaxToki-217M	2 (0.18)	1232	4425	+0.832	+18.4	+0.833	+13.5
MaxToki-217M	4 (0.36)	1232	4425	+0.740	+15.5	+0.735	+11.3
MaxToki-1B	4 (0.20)	2304	4425	+0.766	+18.0	+0.763	+15.1
MaxToki-1B	7 (0.35)	2304	4425	+0.767	+7.2	+0.761	+4.8

Representation	Layer (relative depth)	Width	Genes	EXCESS	Functional- z	EXCESS, shared entries	Functional- z, shared entries
scGPT	2 (0.17)	512	4093	+0.688	+4.0	+0.650	+4.6
scGPT	4 (0.33)	512	4093	+0.813	+3.7	+0.800	+3.3
STATE	3 (0.19)	2048	4761	+0.880	+4.0	+0.863	+7.9
(SE- 600M)							
STATE	6 (0.38)	2048	4761	+0.867	+4.0	+0.853	+6.8
(SE- 600M)							
MaxToki- 217M, untrained	4 (0.36)	1232	4425	+0.441	+4.6	+0.443	+4.1
scGPT, untrained	4 (0.33)	512	4093	+0.296	+2.2	+0.238	-0.7
STATE, untrained	6 (0.38)	2048	4761	+0.761	+1.4	+0.745	+3.8
Expression only, landmarks	—	512	4425	+0.424	-1.3	+0.447	-1.3
Expression only, principal compo- nents	—	256	4425	+0.218	+0.7	+0.227	+0.3
Expression only, centroids	—	50	4425	+0.320	-0.4	+0.328	-0.6

All representations use the same 600 cells per cell type, the same two partitions and the same 5000-gene panel. Model vectors average 50 occurrences and centroid vectors 50 expressing cells per entry; the landmark and principal-component versions use all 300 cells of each half. EXCESS measures how reproducible the gene-specific context shift is; functional-z measures organisation along the nuclear/surface axis against 1000 random-gene axes with non-overlapping poles (no rank regression; genes count-balanced in at least 6 of the 12 cell types). A cell-type pair is scored if it has at least 150 count-balanced genes. This keeps all 66 pairs in the per-model columns and 64 pairs in the last two columns, where only shared entries are used. Table 3 uses at least 200, so EXCESS there can differ by up to 0.003. “Genes” is the number of panel genes count-balanced in at least one cell type; each representation has its own count-balanced entries. The last two columns repeat the analysis on the 14,109 (cell type, gene) entries, from 3957 genes, that are count-balanced in every representation. Relative depth is the number of transformer blocks applied divided by the model’s number of blocks. EXCESS values of different architectures should not be ranked against each other, because width and input type differ (Limitations).

Contextualisation is weakly correlated with what the model gains from context in prediction

If contextualisation serves the autoregressive objective, genes the model contextualises more should gain more from real context in next-gene prediction. We find a weak positive link of this kind. Real context helps prediction a great deal: across 2266 comparisons for 558 genes, the mean benefit is +12.0 nats (median +12.9; a nat is a natural-log unit of probability), so the true gene is roughly 160,000-fold more probable under its real context than under another cell’s. The per-gene size of this benefit is weakly related to how much the gene moves between cell types: the partial Spearman correlation, controlling for how often the gene was sampled

as a target, is +0.161 (95% CI +0.083 to +0.239). Genes whose representations depend more on context tend, weakly, to be genes whose prediction depends more on context.

The link holds in MaxToki-1B and beyond cell identity. In the paired design (the same 350 immune cells, positions and donors for both models, with each gene’s movement taken from the 600-cell extractions), the partial correlation is +0.133 (95% CI +0.040 to +0.222) for MaxToki-217M and +0.200 (+0.108 to +0.280) for MaxToki-1B with a random donor. When the donor comes from the same cell type, real context still helps a great deal (mean benefit +10.83 and +10.36 nats, against +12.07 and +11.53 with a random donor), and the correlation stays positive (+0.236, CI +0.147 to +0.322; +0.323, CI +0.242 to +0.400). It is higher than with a random donor in both models, but we did not test this difference, and the two arms use slightly different genes. So most of the benefit of real context is specific to the cell, not to its cell type, and it is somewhat larger for the genes the model contextualises more. These tests used immune cells only.

Discussion

The results characterise how SCFM representations represent genes in context and what that representation carries. All four models change a gene’s representation with cell type in a gene-specific and reproducible way. In MaxToki-217M, the only model we measured at every layer, this change is most reproducible in the first one or two layers, while genes keep moving between cell types in later layers. This depth pattern is not shared by every model. The other models were measured at only two or three layers: in scGPT, EXCESS rises with depth (+0.687, +0.813 and +0.872 at layers 2, 4 and 9), in STATE it falls slightly (+0.879, +0.867 and +0.842 at layers 3, 6 and 12), and in MaxToki-1B it is the same at layers 4 and 7 (+0.764). Untrained versions of the three architectures keep part of the change (36–88% of the trained EXCESS), so some contextualisation comes from the architecture and the input alone. Training strengthens it. In MaxToki-217M it also strengthens its alignment with functional axes (functional-z +4.6 before training and +15.5 after); in STATE much of that alignment is already there before training. In both MaxToki sizes the change is weakly correlated, gene by gene, with how much the model’s next-gene predictions in immune cells depend on the cell’s own context, beyond its cell type. To our knowledge this is the first measurement for single-cell models of the “how contextual are contextualised representations” question (Ethayarajh 2019). On the nuclear/surface axis, the functional organisation of the change separates MaxToki-217M from scGPT and STATE in every version of the analysis (functional-z +15.5 against +3.7 and +4.0; +16.7 against +1.4 and +5.5 after the rank regression). On the other two axes it does not: on the mitochondrion/cytoskeleton axis STATE is higher after the rank regression (+9.8 against +7.5) but lower with mean expression as the covariate (+7.0 against +9.1), and on the transcription/transport axis the three models are close (+5.0, +3.5 and +4.0). The measure also does not order the other models consistently, so it is not yet a stable way to rank architectures.

What the change carries is organised by co-expression. In no model is the functional organisation significantly stronger than that of gene sets matched to the functional sets on size and average co-expression (smallest $p = 0.07$), and modules of strongly co-expressed genes produce more. This test can detect only large differences, so a small functional effect beyond co-expression cannot be ruled out. Expression alone gives part of this picture but not all of it. A representation that places each gene where it is expressed within each cell type also changes in a gene-specific way (EXCESS +0.321 when each vector averages 50 cells, as in the models, and +0.605 with all expressing cells), so a reproducible, gene-specific change is not by itself a sign of learning. EXCESS values of representations with different widths, inputs and sample sizes should not be ranked against each other. But in none of the three expression-only versions is the change organised along functional axes or along co-expressed gene sets: strong co-expression modules reach 1.4–1.5, 1.5 and 0.9–1.0 times the power of random gene sets, against 5.6–10.1 times in MaxToki-217M, 5.0–8.9 in MaxToki-1B, 3.3–4.2 in scGPT and 12.5–18.0 in STATE. So the models move genes along directions that separate co-expressed gene sets much more than these expression-only versions do, for co-expressed sets in general rather than for functional sets in particular. We do not claim that the models add information beyond co-expression; other expression-only constructions, in particular ones that see co-expression between cell types, might come closer. The two direct tests for context-specific gene function are negative in every model. An expression-only representation can pass the directional test (the centroid build with all expressing cells) and nearly pass the target-activity

test (the centroid build we use; $p = 0.03$ before correction), so a pass would not by itself show function beyond expression. The one signal beyond co-expression is static: in STATE, and more weakly in the larger MaxToki-217M extraction, TFs sit closer to their curated targets than to co-expression-matched non-targets, but not more so in the cell types where those targets are active. Compared with the targets of other TFs, the closeness is not clearly specific to each TF’s own targets in either model (STATE: 17 of 24 TFs positive, $p = 0.032$ before correction; MaxToki-217M: 17 of 26 TFs, $p = 0.084$), so it may partly reflect where regulators and their usual targets sit in general. scGPT TFs are closer to their own targets than to other TFs’ targets, but not closer than co-expression-matched non-targets, so that result is consistent with co-expression alone. In STATE this closeness is already present before training and, if anything, larger; untrained models without a fixed gene table show none. So it most likely comes from the protein-sequence embeddings that serve as STATE’s gene tokens, not from learning on expression data; we did not run STATE with a random gene table, which would test this directly. For MaxToki-217M we cannot say whether the closeness is learned: we did not run an untrained model on the 1000-cell extraction, and on the 600-cell extraction neither the trained nor the untrained model shows it.

The steering results do not test co-expression. Both MaxToki sizes read the nuclear/surface direction more than directions built in the same way from random splits of the same genes. In MaxToki-217M part of this effect goes through its two largest hidden dimensions: with them set to zero, the effect is no longer clearly above the splits ($p = 0.065$). MaxToki-1B also reads the other two functional directions. The random splits mix the two poles, so they are not matched to them on co-expression, and a model that reads co-expressed gene sets would also pass this test. The pushes are about 1.4 to 6 times a gene’s typical (median) natural range along the axis, and each moves half of a cell’s genes at once, so the tests show that the models can read these directions, not that natural context changes along them are used. MaxToki-217M reads the transcription/transport direction in reverse, and its mitochondrion/cytoskeleton direction is not distinct from random splits. The architecture comparison does not contradict the main conclusion. On the nuclear/surface axis STATE organises the change more weakly than MaxToki-217M in every version we ran, and its organisation there disappears when the gene’s mean expression is the covariate (+0.2, against +5.5 with the rank covariate). Against MaxToki-1B the order depends on the gene set: on each model’s own entries STATE is lower (+5.5 against +8.0 after the rank regression), and on the entries shared by all representations it is higher (+6.8 against +4.8). On the other two axes STATE is about as strong as the MaxToki models. scGPT organises the change only weakly. In all four models the organisation stays within the co-expression nulls. The comparison cannot separate input type from the other differences between the models.

One possible reason why we found no context-specific gene function beyond co-expression is that these models are trained only on expression data. They see which genes are expressed and either their rank order (MaxToki) or their expression level (scGPT and STATE). STATE also gets a fixed protein-sequence embedding for each gene. Their training rewards predicting co-expressed genes. The link between contextualisation and the benefit of real context is consistent with that reading, although the test measures only how far a gene moves, not in which direction. Some context-dependent function acts at the protein level, as in moonlighting proteins that switch function with location or binding partners (Jeffery 1999), and need not change expression. Other forms, such as cell-type-specific enhancer choice by lineage-determining TFs (Heinz et al. 2010), act through target-gene expression and could in principle be visible; our tests did not detect them. STATE’s protein-sequence gene tokens are the same in every cell type, so they cannot carry a gene’s changed role by themselves. We did not test multi-omic models, models that read regulatory DNA sequence together with expression, or much larger expression models.

Limitations

The depth of analysis differs between models. The rank control, the co-expression and tightness nulls, both Level 2 tests and the expression-only comparison were run on the four trained models and three expression-only representations, on the same 600 cells per cell type. The untrained models went through EXCESS, functional-z against random axes and the curated-target test; untrained scGPT and STATE also went through the rank regression on the functional axes. Each untrained model was built once from one random seed, and we built an untrained MaxToki-217M only, not an untrained MaxToki-1B. The Setty replication, the full

depth profile and the 1000-cell analyses were run on MaxToki-217M only; on the Setty data we computed only EXCESS and functional-z against random axes. Steering and the prediction link were run on the two MaxToki models only, and the cells that were pushed or scored were immune cells (30 for steering, 350 for the prediction link). We did not steer scGPT or STATE: neither has a next-gene readout comparable to MaxToki's, and their expression heads would first need a check that they track the input, which we did not do. With 600 cells per cell type fewer genes reach the occurrence cap in many cell types, and only 11 TFs qualified for the curated-target test in the MaxToki models and the expression-only representations (16 for scGPT, 24 for STATE), so that test has little power for those representations. The Level 2 tests were run only with genes count-balanced in at least 6 cell types. Each model has its own count-balanced entries; on the entries shared by all representations EXCESS barely changes, but the functional-z order changes for STATE, scGPT and MaxToki-1B, and only EXCESS and nuclear/surface functional-z were repeated on the shared entries. Functional-z also moves with the draw of random axes, so only clear differences between models should be read as real. EXCESS values should not be ranked across architectures, because widths and input types differ. Input lengths also differ: MaxToki saw each cell's top 1022 genes by count divided by gene median, scGPT a random 1199 of its expressed genes and STATE the top 2047 by count. scGPT and STATE receive each gene's expression level; the rank regression used MaxToki's mean rank on the same cells. Repeating it with mean log-normalised expression left every co-expression-null result and MaxToki's organisation unchanged but removed STATE's nuclear/surface organisation (Additional file 1); we did not repeat it for the EXCESS rank regression, the tightness null or the two Level 2 tests.

The matched co-expression null is wide: depending on the model, an axis would need about 1.5 to 2.9 times the matched-set mean to reach $p < 0.05$, so this test could detect only a large excess. It also matches only average pairwise co-expression; power depends on how co-expression is distributed within a gene set. Two of the three expression-only representations use correlations within each cell type and barely respond to co-expression. The third is built on axes from all cell types pooled, but each vector is still an average over cells of one cell type, so it may not capture co-expression between cell types either; none of the three responds to co-expressed gene sets. So the contrast between the models and these versions does not show that the models add anything beyond co-expression in the data. The third version's result also depends on how it is built: with 512 instead of 50 principal components it failed the positive control, and we did not run other numbers of components through the pipeline. Other constructions, such as gene embeddings learned from the same cells or a co-expression network embedding, were not tested and might reproduce more of the organisation. The rank control rests on regressions and a rank-matched comparison; it cannot exclude every way in which position could enter a causal model. Most analyses use one layer per model, at matched relative depth, not the peak layers 1–2 of MaxToki-217M; the other models were not measured at every layer. The steering direction is the context-averaged difference between two gene classes, not the context-specific change itself, and the pushes are large; we did not steer along directions built from gene sets matched to the poles on co-expression, so steering cannot separate function from co-expression; and with 30 random splits, the split-control p values cannot survive a correction for several tests. The prediction link is a correlation across genes, and it controls only for how often a gene was sampled as a target. A gene whose expression varies more from cell to cell could both move more between cell types and gain more from its own context, whatever the model does with that movement. We did not control for this, for example with the gene's rank change or the spread of its expression, and we did not check whether movement measured in the untrained models or the expression-only representations shows the same link. The Setty data cover a pre-filtered set of genes, unlike the full raw counts of Tabula Sapiens. Cell type is partly tied to tissue and donor. Of the 12 cell types, 5 come from one dataset only: 4 from lung, and kidney epithelial cells, all from a single donor. Most pulmonary alveolar type 1 cells (915 of 1000) also come from one donor. Assay mix differs less: in every cell type at least 86% of cells are 10x 3' v3. The two halves of a cell type share its tissue, donor and ambient-RNA background, so differences in these between cell types count toward EXCESS and repeat in both halves. We did not measure how large this part is, for example by comparing cell types from the same tissue and donor. In the Setty data all contexts come from one bone-marrow dataset and EXCESS is similar (+0.713), so the effect does not need these differences, but their share in the Tabula Sapiens numbers is unknown. The Level 2 results are negative, but they do not show that context-specific gene function does not exist. Some of it leaves no trace in expression, and the directional test could detect only clear congruence, so weaker effects could be

missed. Functional axes are built from direct annotations in one annotation file; the conclusions hold across three axes, but specific z-scores depend on how the poles are defined.

Conclusions

The four models we tested build contextual gene representations that are worth studying in their own right. A gene's vector changes with its cell type in a gene-specific, reproducible way. Part of this change comes from the architecture and part from training, and a representation built from expression alone also changes in a gene-specific way. In MaxToki-217M, the only model scanned layer by layer, it is most reproducible in the first one or two layers; scGPT, measured at three layers, shows the opposite trend. In both MaxToki sizes, genes that change more are, weakly, the genes whose next-gene prediction gains more from the cell's own context. How the change is organised differs between architectures. The organisation follows co-expression: in four models, context does not move genes significantly more along functional axes than along axes built from gene sets with the same co-expression. Context moves genes along axes built from co-expressed gene sets much more in the models than in three expression-only representations of the same cells. Among the things we measured, this is what separates the models from expression alone. Two direct tests for context-specific gene function are negative in every model. So we could not read a gene's changed role in a new cell type from these models, which is the use that would make them a tool for new hypotheses. Our tests could detect only clear effects, so this does not prove the models lack it, and whether models trained on more than expression can show it was not tested.

List of abbreviations

AUC: area under the receiver operating characteristic curve; CI: confidence interval; ESM2: Evolutionary Scale Modeling 2; GO: Gene Ontology; PC: principal component; ROC: receiver operating characteristic; SCFM: single-cell foundation model; TF: transcription factor; UCE: Universal Cell Embeddings.

Declarations

Ethics approval and consent to participate

Ethics approval and consent to participate were not required for this study, which is a secondary analysis of publicly available, de-identified human single-cell RNA-sequencing data (Tabula Sapiens (Tabula Sapiens Consortium 2022) and Setty et al. (Setty et al. 2019)) and involved no contact with participants. The original studies obtained ethics approval and informed consent, as described in those publications. No new human data or samples were collected.

Consent for publication

Not applicable.

Availability of data and materials

The datasets analysed during the current study are available in the following public repositories: the Tabula Sapiens immune, kidney and lung datasets in CZ CELLxGENE Discover (Tabula Sapiens Consortium, n.d.-a, n.d.-b, n.d.-c); the human CD34+ bone-marrow data of Setty et al. (Setty et al. 2019), as distributed with scVelo, in figshare (scVelo project, n.d.); the MaxToki checkpoints in Hugging Face (Theodoris Lab 2026); the scGPT whole-human checkpoint (Bo Wang Lab, n.d.); the STATE SE-600M model in Hugging Face (Arc Institute 2025); the Geneformer gene dictionaries in Hugging Face (Theodoris, n.d.); the GEARS gene-to-GO annotation file in Harvard Dataverse (Roohani et al. 2022); TRRUST v2 (Han et al. 2018a); and DoRotheA (Saez-Rodriguez Lab, n.d.). All analysis code, the result files from which every number, table and figure is computed, the figure code and the list of the 20,000 immune cells used are available in the gene-context

repository (Kendiukhov 2026a) and archived at Zenodo (DOI 10.5281/zenodo.23119961) (**web_zenodo?**). The representation tensors for MaxToki-217M, MaxToki-1B, scGPT, STATE, the three untrained models, the three expression-only representations and the Setty data (about 10 GB) are not deposited because of their size. They can be regenerated from the public inputs with the provided extraction scripts and are available from the corresponding author on reasonable request. No new experimental data were generated.

Software availability:

- Project name: gene-context
- Project home page: <https://github.com/Biodyn-AI/gene-context>
- Archived version: Zenodo, DOI 10.5281/zenodo.23119961 (release `release2` of the project repository, the same code as tag `bmc-submission-v3.1`) (**web_zenodo?**)
- Operating system(s): platform independent (developed and run on macOS, Apple silicon)
- Programming language: Python 3.12
- Other requirements: NumPy, SciPy, scikit-learn, h5py, matplotlib and openpyxl; for the extraction scripts, PyTorch (2.2 or later) and transformers (4.43 or later); for scGPT, a local copy of the scGPT source code and the whole-human checkpoint (Bo Wang Lab, n.d.) (only its model file is imported) and tqdm; for STATE, the Arc Institute `state` package and `anndata`; versions used: PyTorch 2.12.1, transformers 5.13.0, NumPy 2.4
- License: MIT
- Any restrictions to use by non-academics: none

Competing interests

The author declares that he has no competing interests.

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Authors' contributions

IK conceived and designed the study, wrote the software, carried out the analyses, interpreted the results and wrote the manuscript. The author read and approved the final manuscript.

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Figures

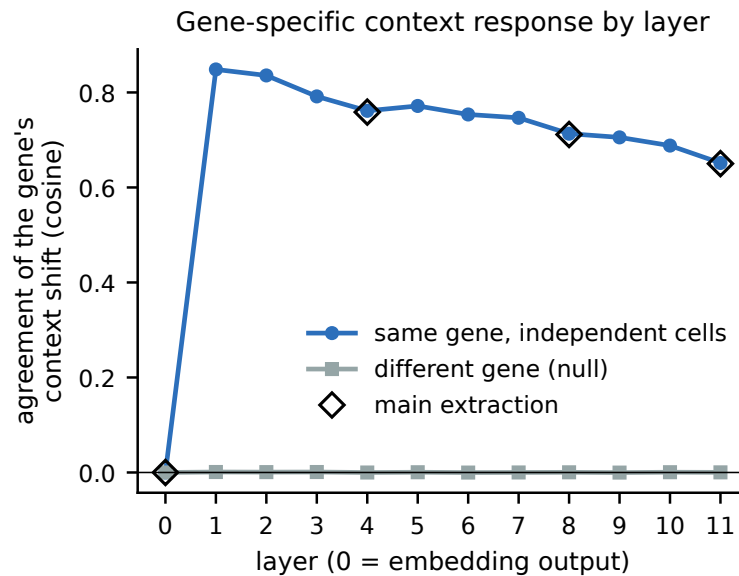


Figure 1: **Figure 1. Gene-specific context response from the embedding through all 11 layers of MaxToki-217M.** Agreement of the same gene’s relative context shift across independent halves of the cells (blue; full depth scan with a 4000-gene panel) against the gene-shuffled null (grey). Open diamonds mark the four layers of the main 6000-gene extraction (Table 1). Layer 0, the context-free embedding output, is exactly zero; the response is highest at layer 1 (+0.847) and falls to +0.651 at layer 11. Values are pooled over 66 cell-type pairs per layer; gene-level bootstrap 95% CIs are narrower than the markers.

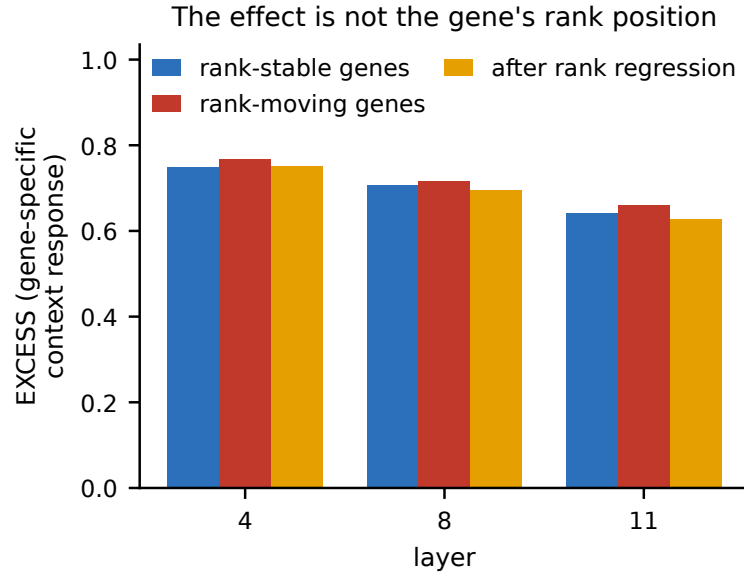


Figure 2: Figure 2. The gene-specific context response is not explained by rank position. EXCESS (the agreement, as a cosine, of a gene’s relative context shift between two independent halves of the cells, minus the same agreement for a randomly paired gene; Table 1) in MaxToki-217M (main extraction, 1000 cells per cell type) at layers 4, 8 and 11 for genes whose rank changes little between cell types (rank-stable, below the median rank change), genes whose rank changes a lot (rank-moving, above the median), and all genes after a linear regression of the shift on the gene’s ranks and rank change. The response is about the same for rank-stable and rank-moving genes, and 99%, 98% and 97% of it survives the rank regression at layers 4, 8 and 11 (a regression with curved rank terms gives almost the same; Results).

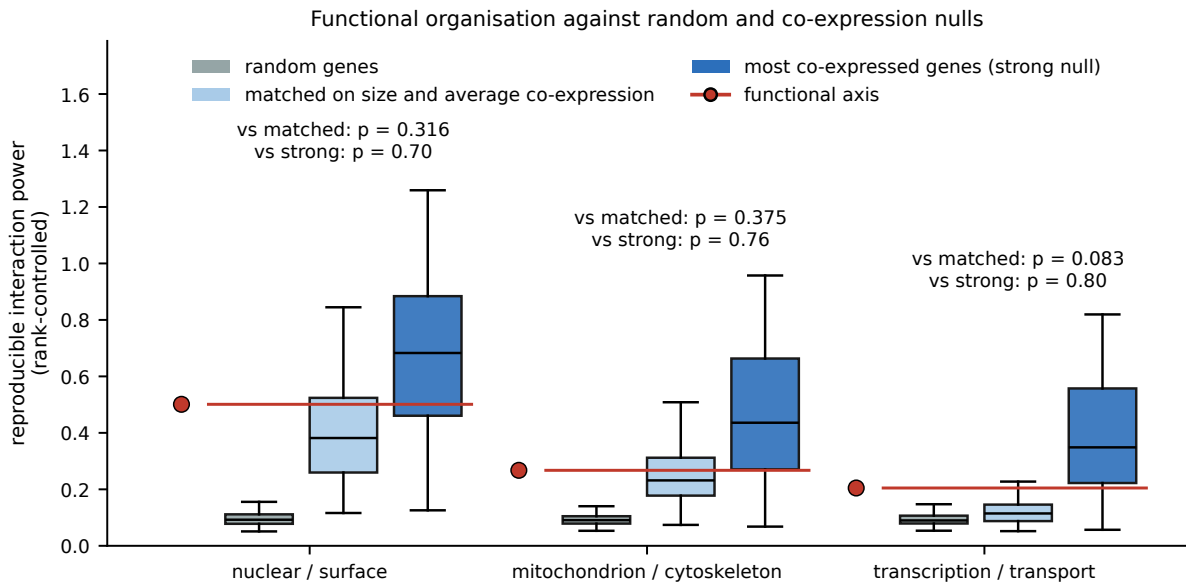


Figure 3: Figure 3. Functional organisation compared with random and co-expression nulls. For each of the three functional axes with well-separated poles (MaxToki-217M, main extraction with 1000 cells per cell type, layer 4; the cell-cycle/differentiation axis is omitted), the red marker and line show the functional axis’s rank-controlled reproducible interaction power. Boxes show 300 null axes each, all with the functional poles’ sizes: random genes (grey); gene sets matched on size and average co-expression (light blue); and modules of the genes most co-expressed with a random seed gene (dark blue), which are 3–11 times more co-expressed than the functional poles. Boxes show the interquartile range and median; whiskers reach the most extreme null value within 1.5 times the interquartile range (values beyond are not drawn). Annotated p values are empirical one-sided values against the matched null and against the strongly co-expressed modules. The functional axes are far above random axes but within both co-expression nulls.

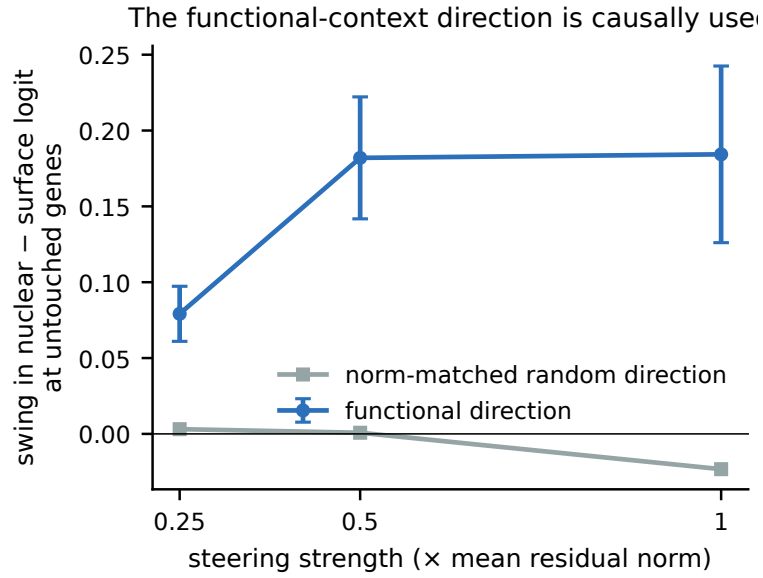


Figure 4: **Figure 4. The nuclear/surface direction is read by the model.** We added the nuclear/surface direction at the layer-4 representation of half of each cell’s genes (MaxToki-217M, 30 immune cells) and read the model’s logits at the other, untouched half. The y axis is the signed swing: the specificity (the change in mean logit of nuclear-pole genes minus that of surface-pole genes) after a push toward the nuclear pole, minus the specificity after a push toward the surface pole. The x axis is steering strength, in multiples of the mean residual-stream norm. Blue: functional direction (mean $\pm$ standard error over 30 cells). Grey: random direction of the same norm, drawn fresh for each cell. Only the nuclear/surface direction is shown. In the same test, the mitochondrion/cytoskeleton direction has no significant effect at strengths 0.25 and 0.5, and the transcription/transport direction has a reversed effect (Results).

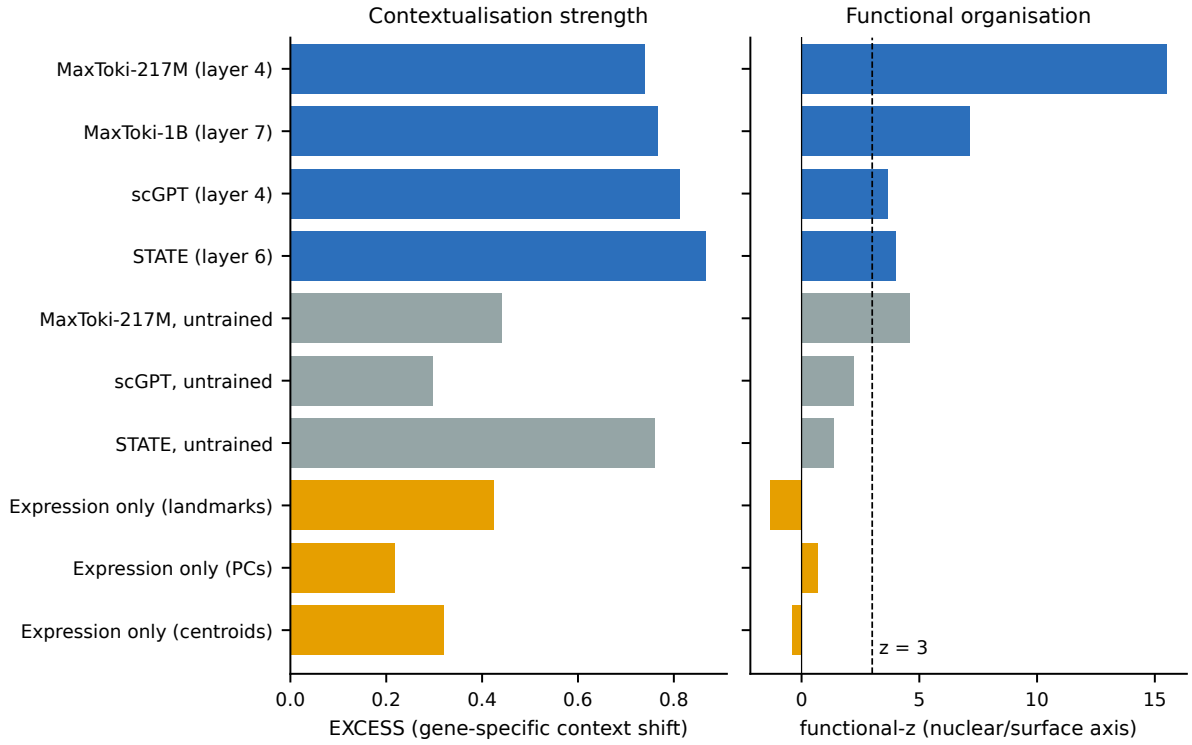


Figure 5: **Figure 5. Contextualisation strength and functional organisation of every representation on the same cells.** Left: EXCESS (agreement of the same gene’s context shift between two halves of the cells, minus that for a randomly paired gene; as in Fig. 2); values of different architectures should not be ranked against each other, because widths and input types differ. Right: functional-z for the nuclear/surface axis against 1000 random-gene axes (no rank regression; genes count-balanced in at least 6 of the 12 cell types). Blue: trained models at depth-matched layers (MaxToki-217M layer 4, MaxToki-1B layer 7, scGPT layer 4, STATE layer 6). Grey: untrained versions of the same architectures (the untrained STATE uses the same fixed protein-embedding gene table as the trained model; only its other weights are random). Orange: expression-only representations built from the same cells without a model. All use the same 600 cells per cell type, partitions and 5000-gene panel; model and centroid vectors average 50 occurrences or expressing cells per entry, and the landmark and principal-component versions all 300 cells of each half. The dashed line marks $z = 3$.